

Rotational Dynamics

Rotational analog for:

→ inertia

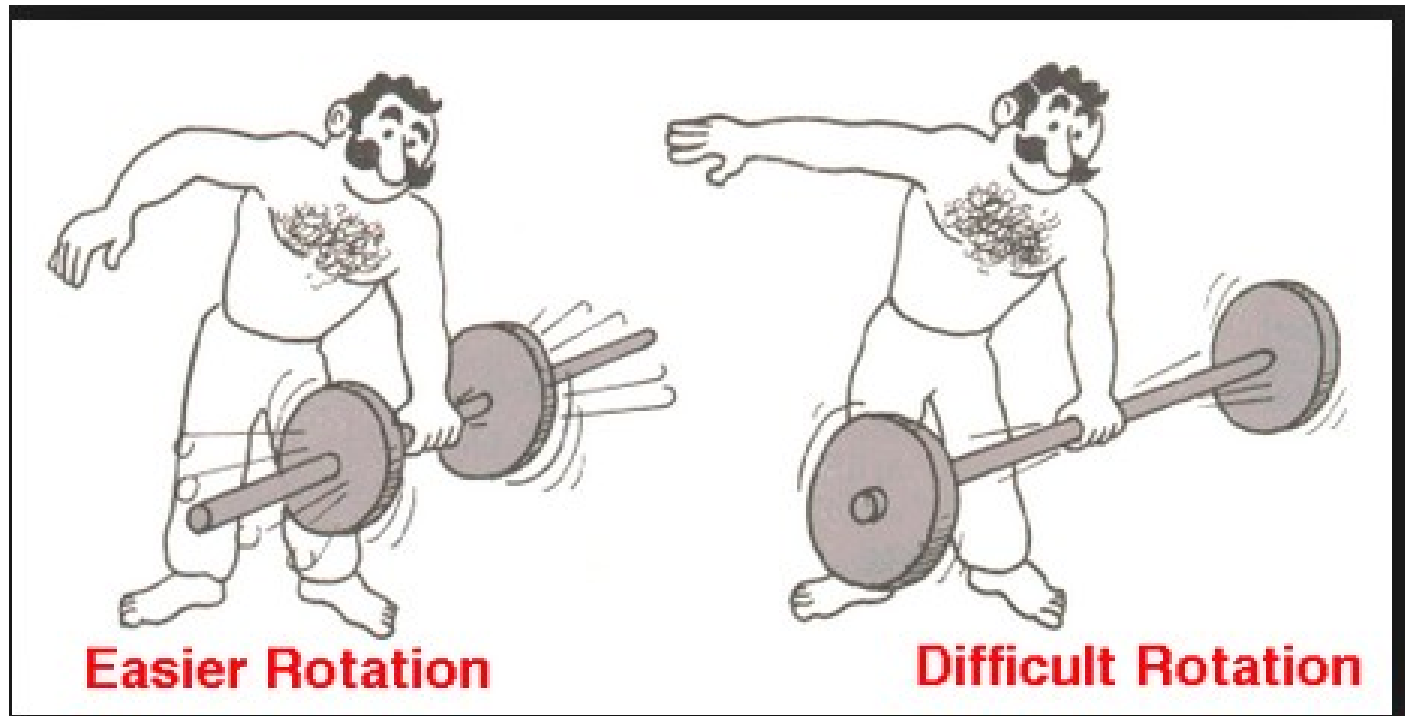
→ Newton's 2nd law

→ Kinetic energy

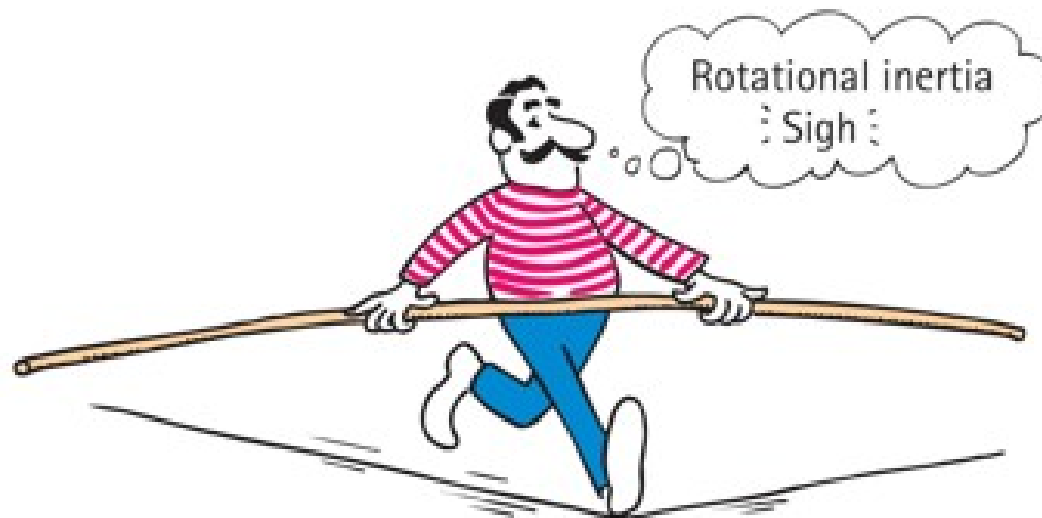
→ Angular momentum

Rotational analog for mass moment of inertia

Rotational inertia (I): resistance to changes in rotational motion



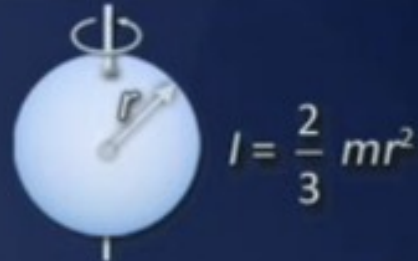
- The greater the rotational inertia, the harder it is to change its rotational state.
 - A tightrope walker carries a long pole that has a high rotational inertia, so it does not easily rotate.
 - Keeps the tightrope walker stable.



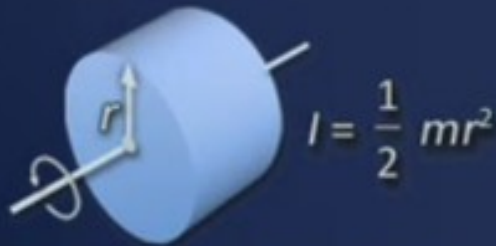
Rotational Inertias



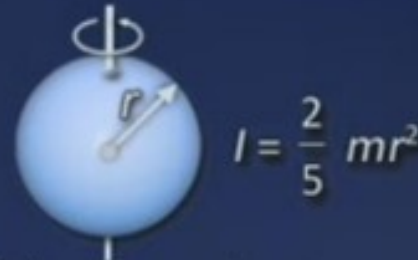
Thin ring or hollow cylinder about its axis



Hollow spherical shell about diameter

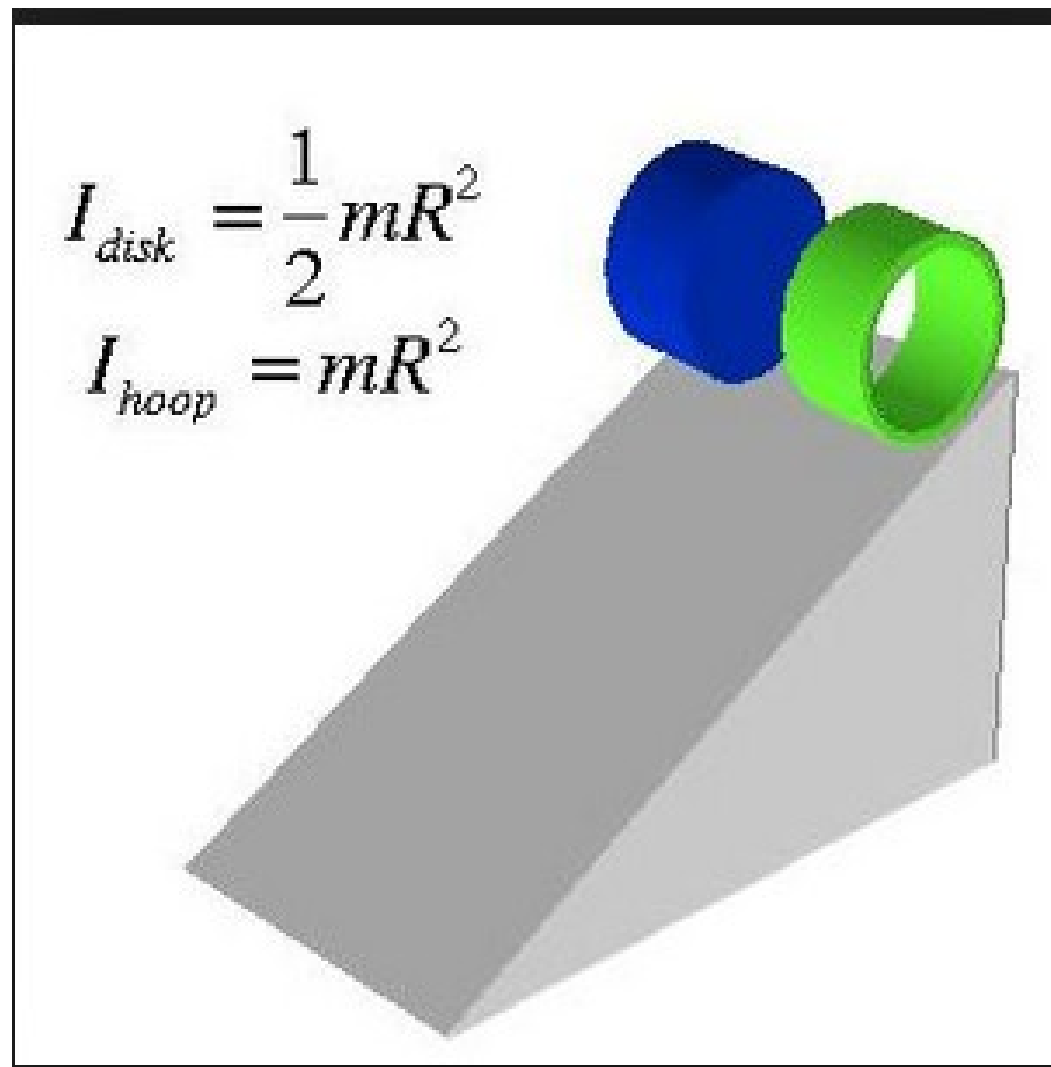


Disk or solid cylinder about its axis



Solid sphere about diameter

Moment of inertia/rotational depends on the DISTRIBUTION of mass around the axis of rotation.



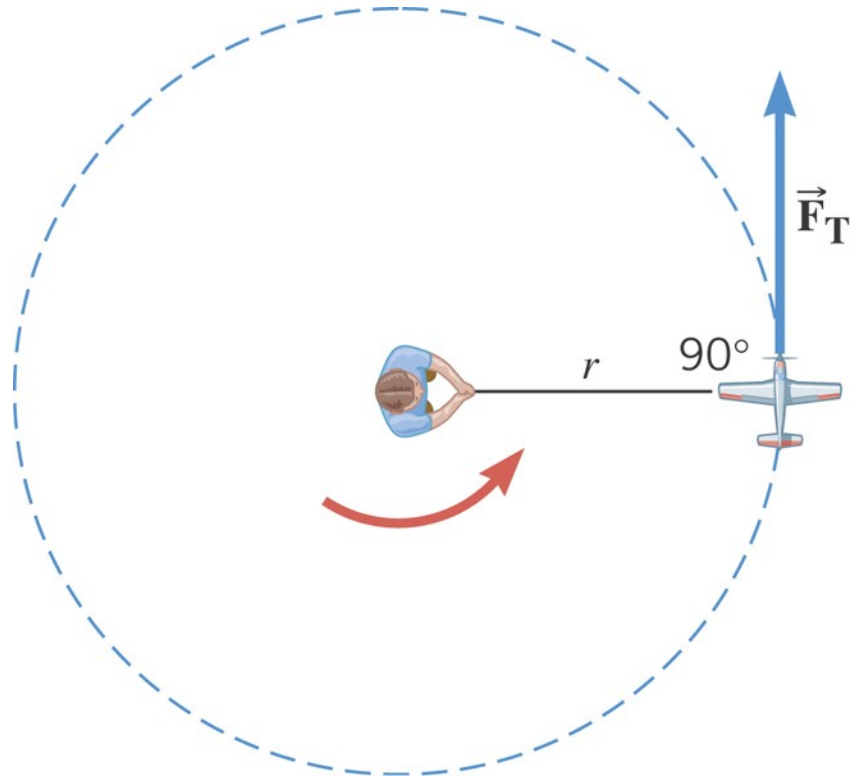
Movie race of the objects.

Its “falling “ so mass does not matter. Interestingly, R does not matter.
We can find the acceleration using the conservation of energy.

Rotational analog for mass = moment of inertia

$$I = m r^2$$

For a point mass

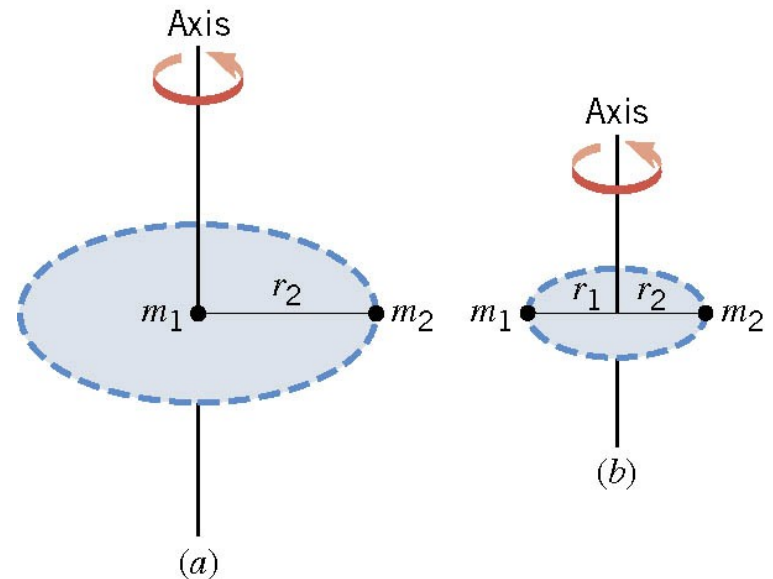


Moment of Inertia, I

9.4 Newton's Second Law for Rotational Motion About a Fixed Axis

Example 9 The Moment of Inertia Depends on Where the Axis Is.

Two particles each have mass and are fixed at the ends of a thin rigid rod. The length of the rod is L . Find the moment of inertia when this object rotates relative to an axis that is perpendicular to the rod at (a) one end and (b) the center.



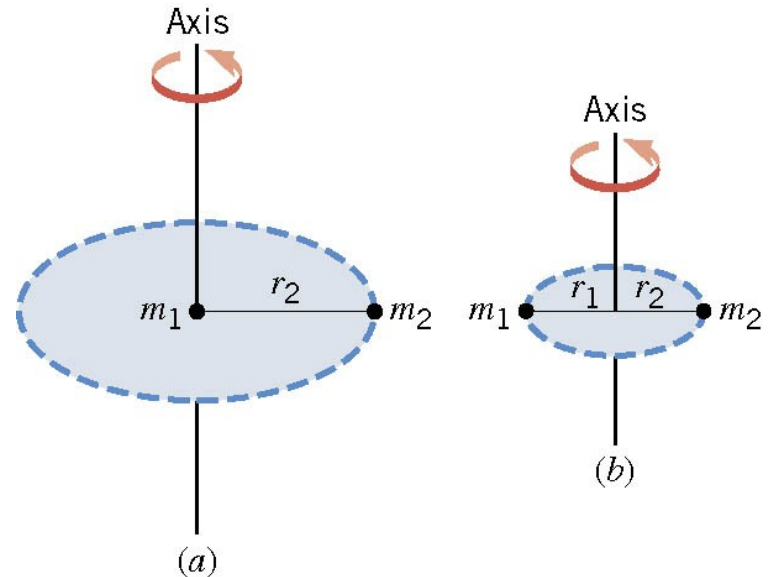
9.4 Newton's Second Law for Rotational Motion About a Fixed Axis

$$(a) \quad I = \sum (mr^2) = m_1 r_1^2 + m_2 r_2^2 = m(0)^2 + m(L)^2$$

$$m_1 = m_2 = m$$

$$r_1 = 0 \quad r_2 = L$$

$$I = mL^2$$



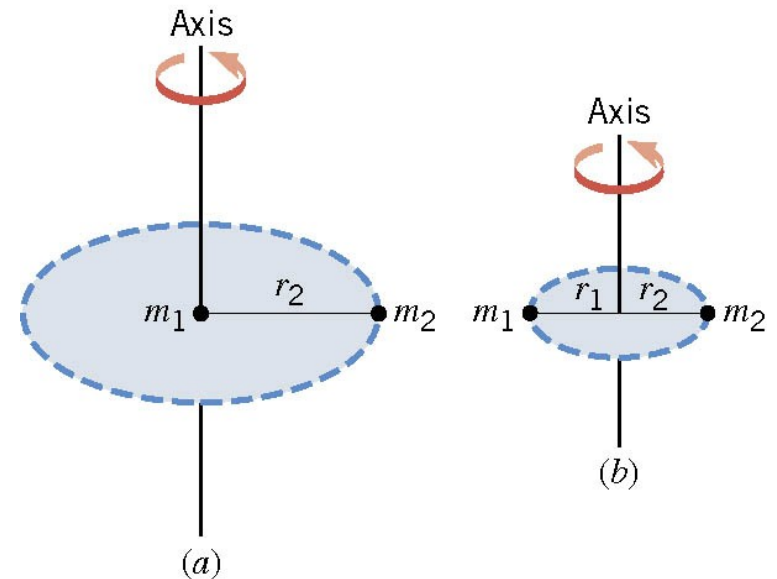
9.4 Newton's Second Law for Rotational Motion About a Fixed Axis

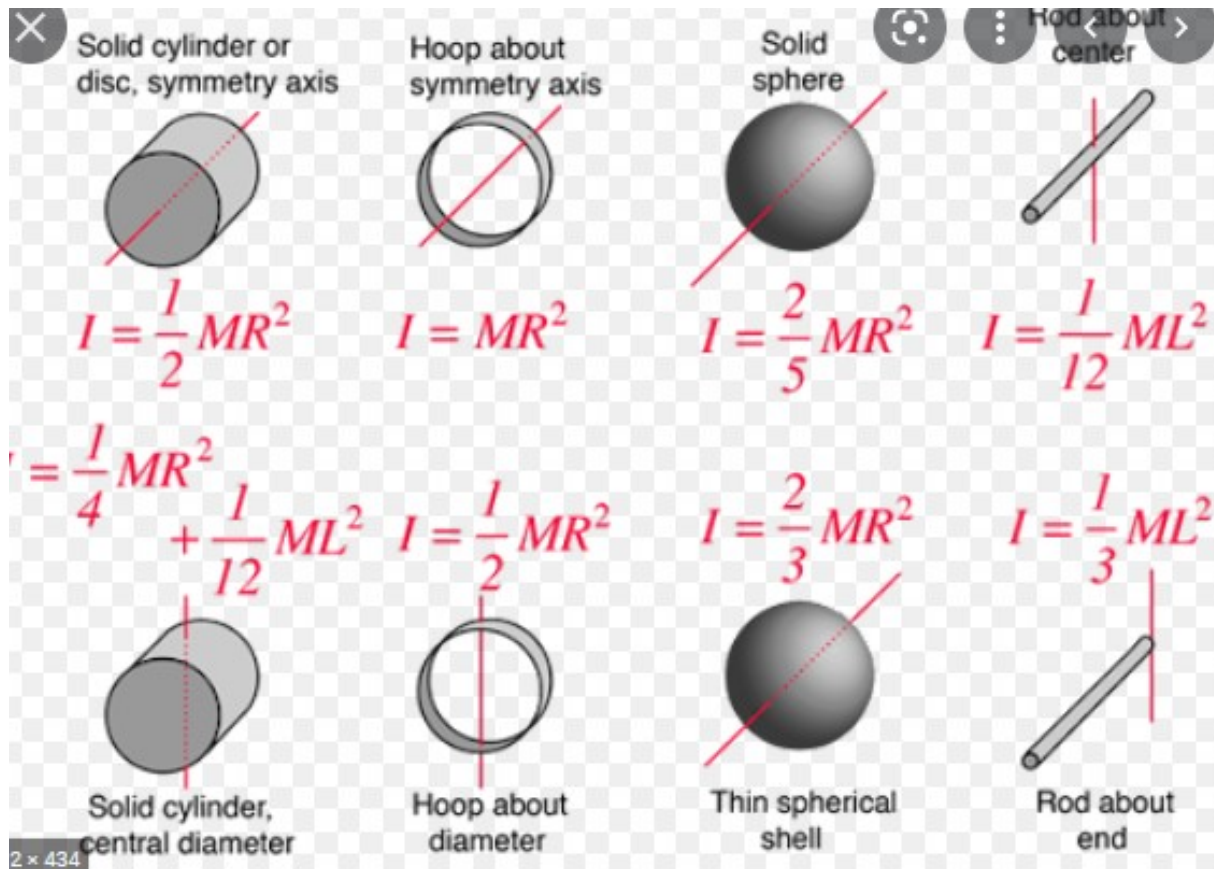
$$(b) \quad I = \sum (mr^2) = m_1 r_1^2 + m_2 r_2^2 = m(L/2)^2 + m(L/2)^2$$

$$m_1 = m_2 = m$$

$$r_1 = L/2 \quad r_2 = L/2$$

$$I = \frac{1}{2} mL^2$$

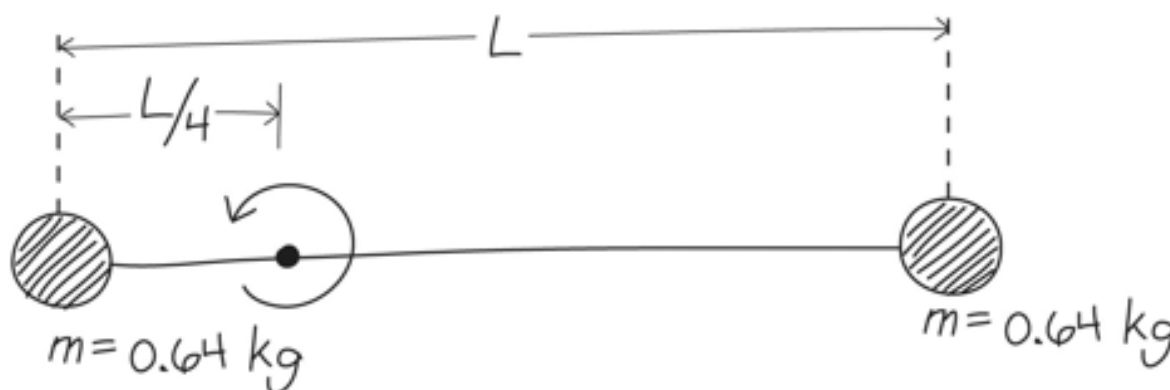




Clicker Question 2



- Consider the dumbbell in the figure. How would its rotational inertia change if the rotation axis were at the center of the rod?
- A. I would increase.
- B. I would decrease.
- C. I would remain the same.



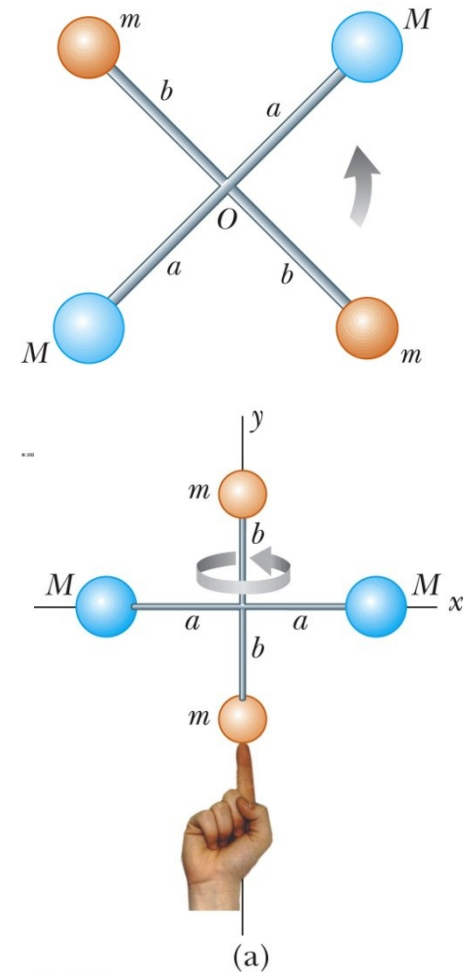
Clicker Question 8

- Two uniform solid spheres have the same radius, but one is three times more massive than the other. What is the ratio of the larger moment of inertia to that of the smaller moment of inertia?

- A. 3
- B. 6
- C. 9
- D. 16
- E. 27

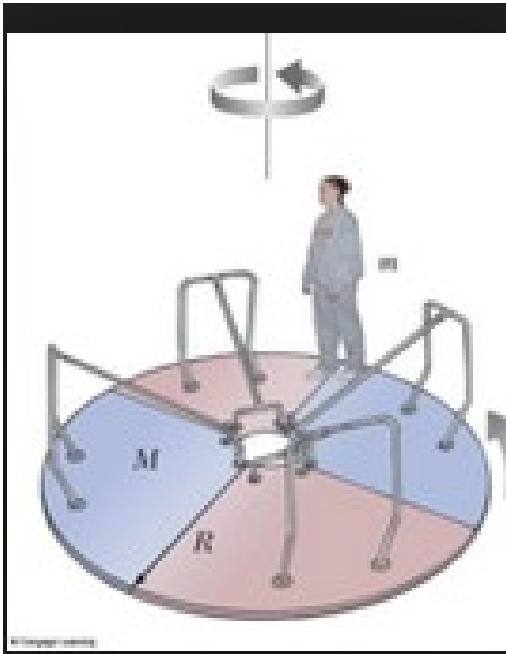
The Baton Twirler

- Consider an unusual baton made up of four spheres fastened to the ends of very light rods. Each rod is 1.0m long ($a = b = 1.0$ m). $M = 0.3$ kg and $m = 0.2$ kg.
- (a) Find I about an axis perpendicular to the page and passing through the point where the rods cross.
- (b) The majorette tries spinning her strange baton about the axis y , calculate I of the baton about this axis

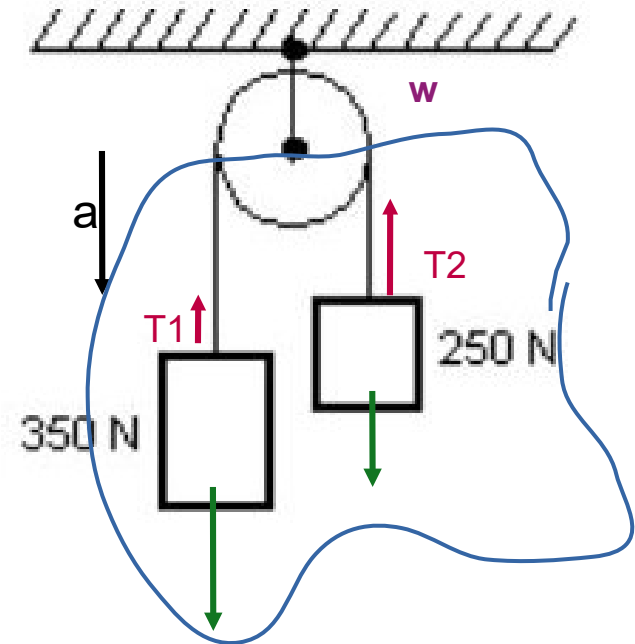


Textbook 9.6

The moment of inertia of a rigid composite system is the sum of the moments of inertia of its component subsystems (all taken about the same axis).



The mass of the person is 60Kg
 The mass of the rotating platform is 500Kg
 The radius is 2 m
 What is the total inertia?



Consider the system =
 Pulley of mass M , radius R
 (plain cylinder) and the 2
 masses 25kg and 35 kg.
 Its like the masses are glue to
 The pulley (forget about the
 string). What is the total inertia?

Rotational dynamics.

There is a symmetry with linear dynamics.

Newton's 2nd law: $F = ma$

Rotational analog of Newton's 2nd law: $\tau = I\alpha$

Kinetic energy: $K = \frac{1}{2}mv^2$

Rotational kinetic energy: $K_{\text{rot}} = \frac{1}{2}I\omega^2$

Momentum: $p = mv$

Angular momentum: $L = I\omega$

Analog for Newton's 2nd law

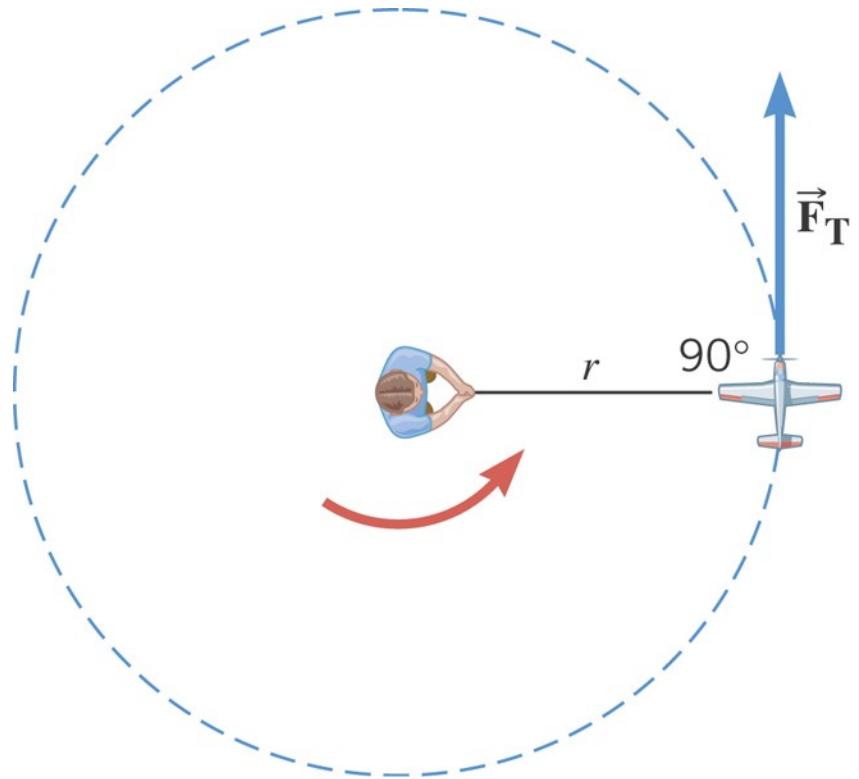
$$F_T = m a_T$$

$$a_T = r \alpha$$

$$\tau = F_T r$$

$$\tau = (m r^2) \alpha$$

Moment of Inertia, I



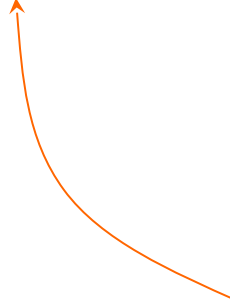
9.4 *Newton's Second Law for Rotational Motion About a Fixed Axis*

ROTATIONAL ANALOG OF NEWTON'S SECOND LAW FOR A RIGID BODY ROTATING ABOUT A FIXED AXIS

Net external torque/axis = (moment of inertia/axis) x (rotational acceleration)

$$\sum \tau = I \alpha$$

Requirement: Angular acceleration must be expressed in radians/s².


$$I = \sum (mr^2)$$



<https://www.youtube.com/watch?v=EixtGKNP6ok>

028 VIDEO in folder

The angular speed are equal

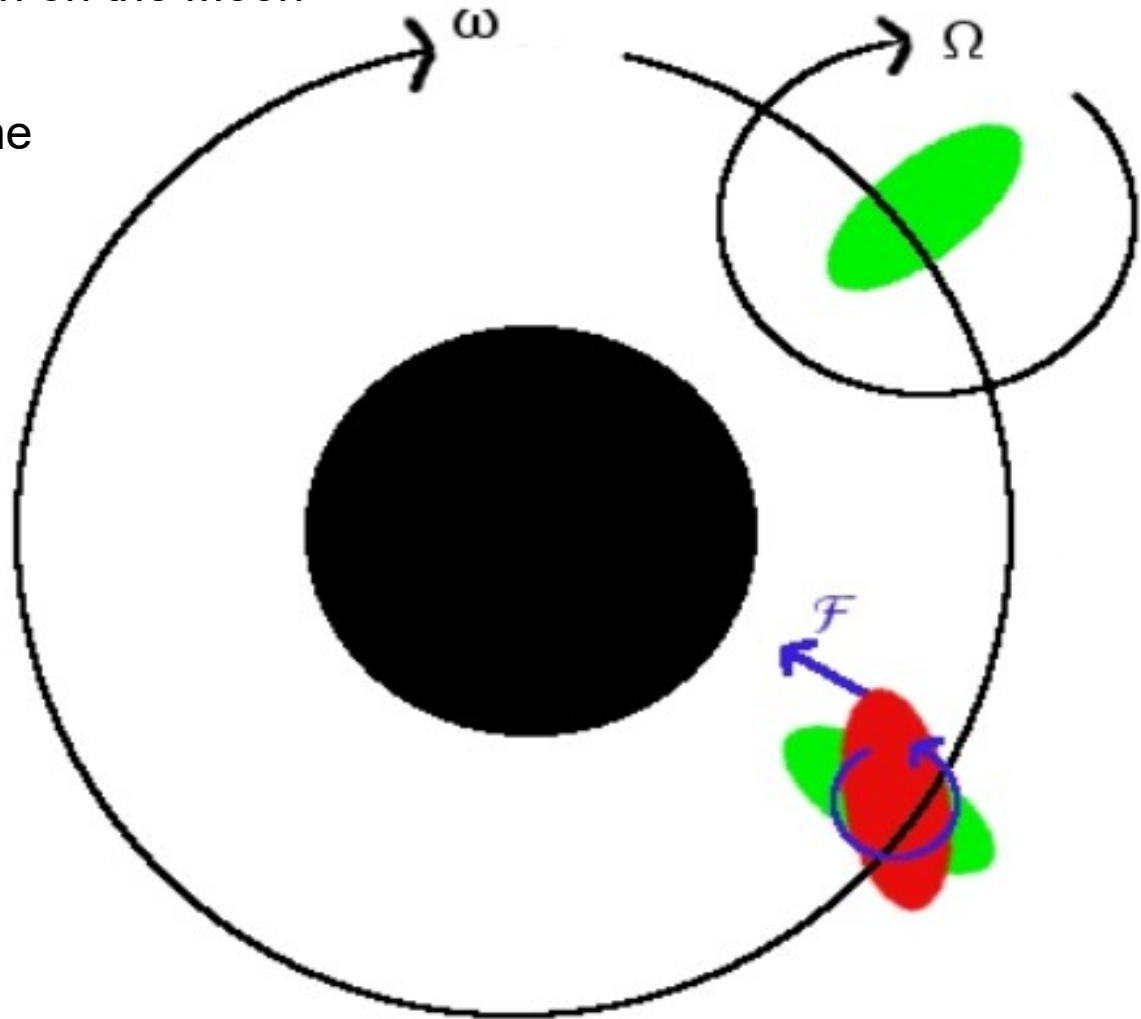
The torque from the Earth on the Moon

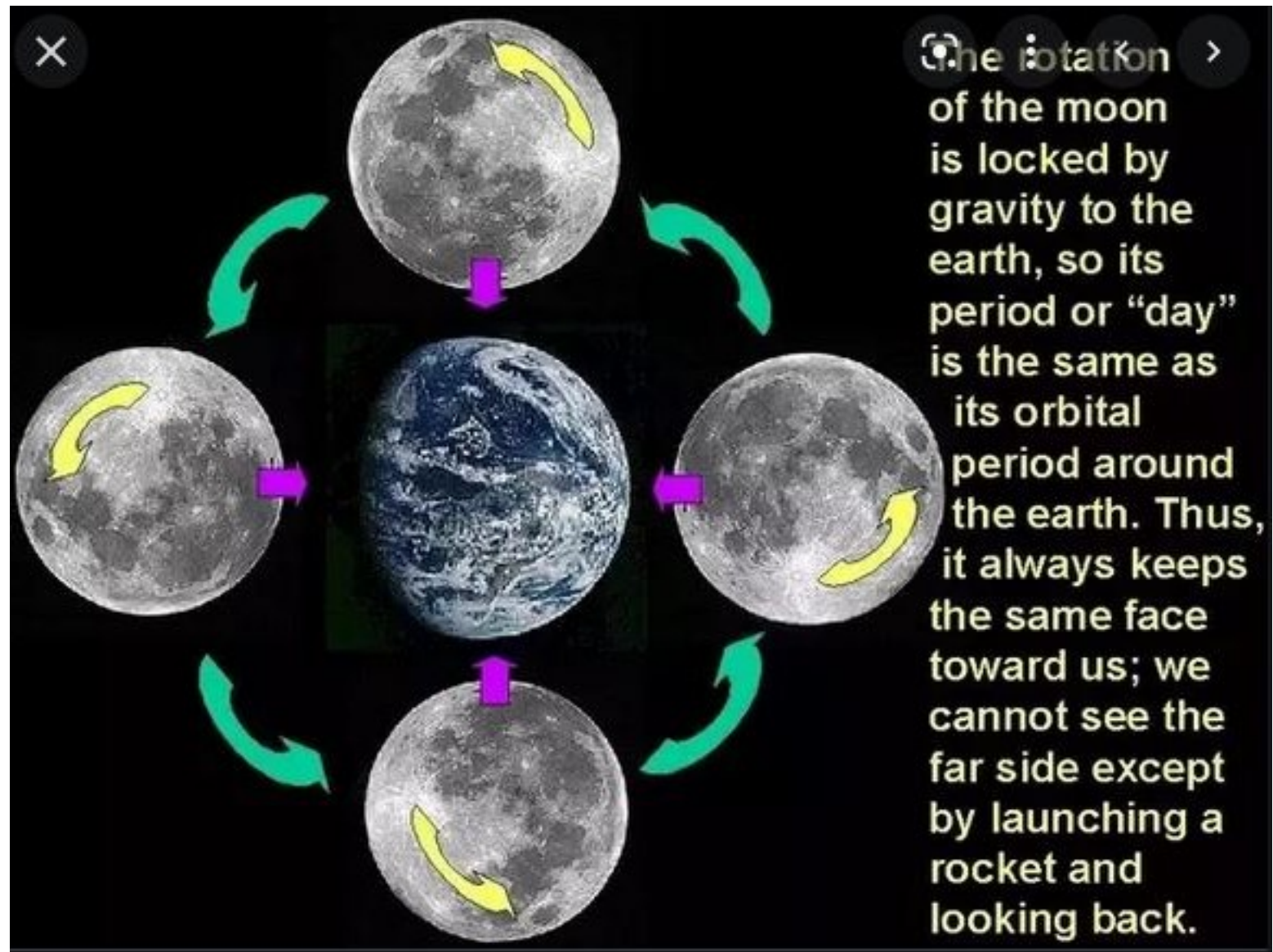
Locks the Moon.

We always sees the same
Face.

The Earth and the Moon
are locked together

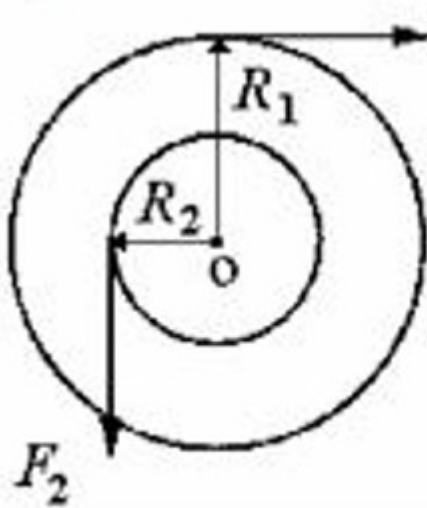
Same period for
Rotation around the Earth
Or for the rotation of
The Moon about its axis.





The rotation of the moon is locked by gravity to the earth, so its period or "day" is the same as its orbital period around the earth. Thus, it always keeps the same face toward us; we cannot see the far side except by launching a rocket and looking back.

Example: wheel and axle → its a machine



Write Newton's second law

Sum(torque) = Inertia x acceleration

$F_1 R_1 - F_2 R_2 = I \times \text{acceleration}$

I is the inertia of the pulley (cylinder)

If equilibrium → $F_1 R_1 - F_2 R_2 = 0$

Say F_1 is the effort and F_2 is the load and we are pulling at a constant speed.

What is the mechanical advantage ? (F_1/F_2 ?)

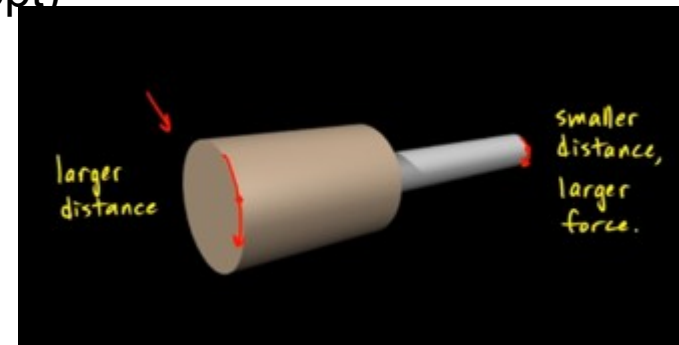
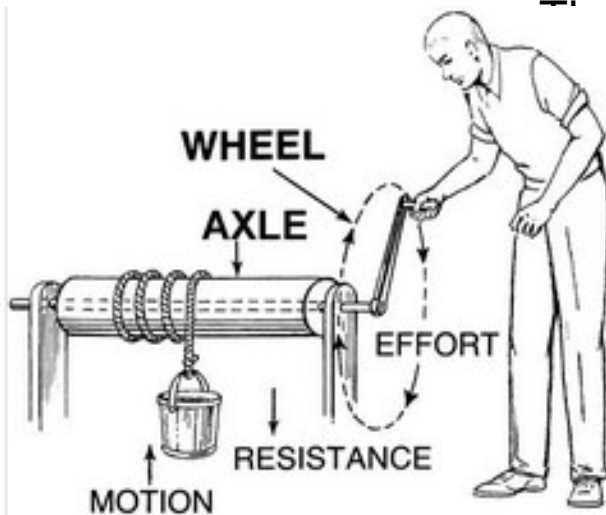
Big radius x small force = small radius x big force

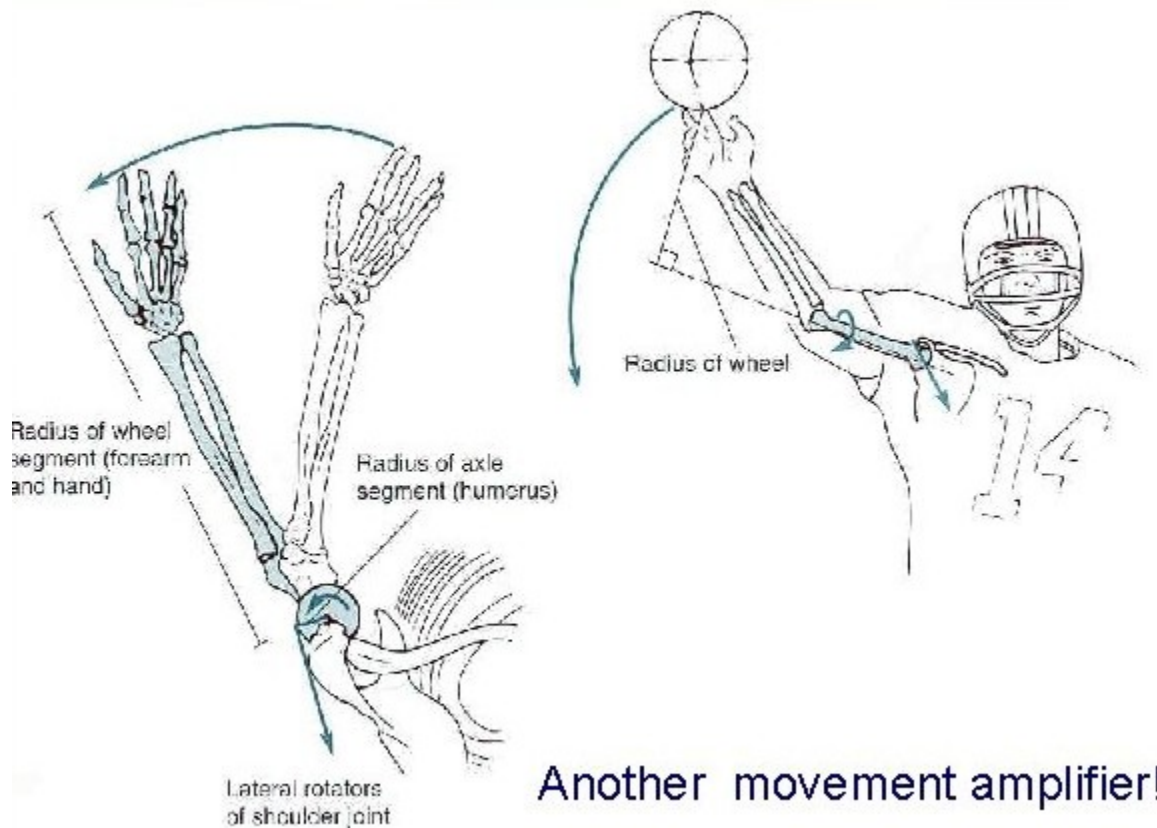
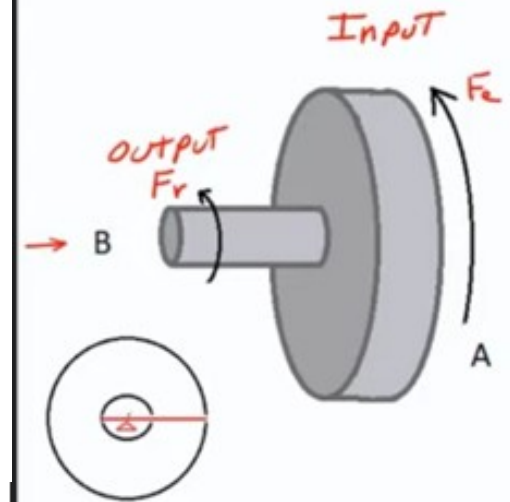
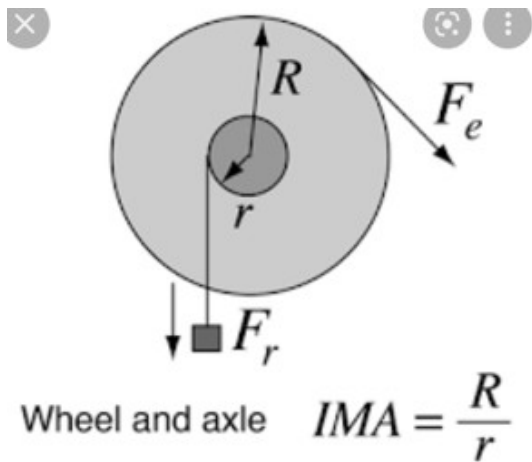
it means: (same angle swept)

distance x small effort = small distance x large load

(distance = radius x angle swept)

conservation of energy



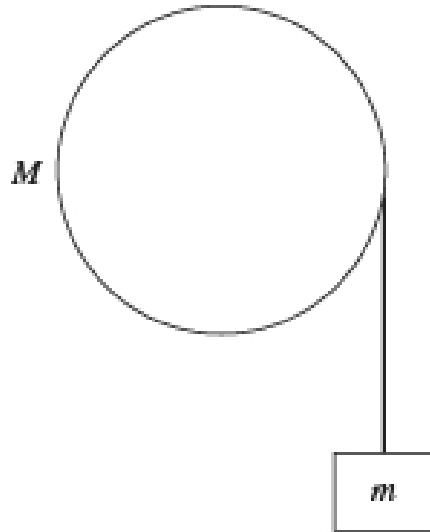


The MA is smaller than one.
The muscle applies a huge Force over small distance (small contraction)

That translates to large Motion but small force.

The lower arm bone Is light. (small load)
But fragile. Easy to break.

- 3) A cinder block of mass $m = 4.0 \text{ kg}$ is hung from a nylon string that is wrapped around a frictionless pulley having the shape of a cylindrical shell, as shown in the figure. If the cinder block accelerates downward at 4.90 m/s^2 when it is released, what is the mass M of the pulley?

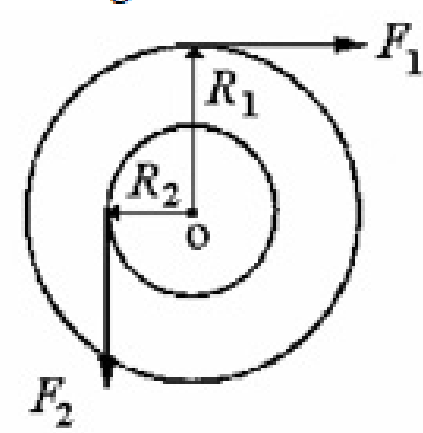


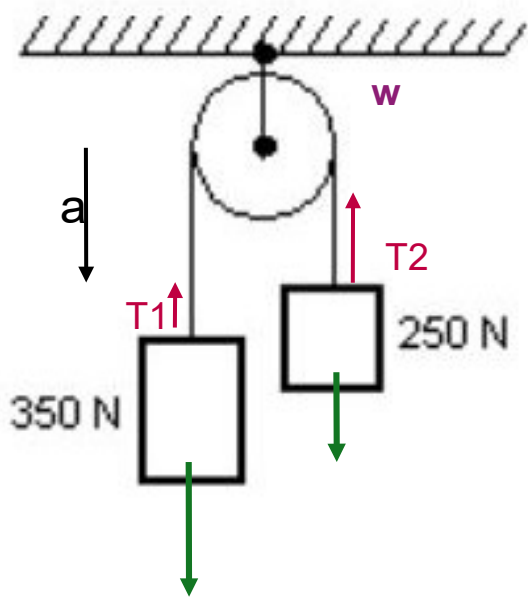
Use the relationship (no slipping of the rope)
Linear acceleration = R (angular acceleration)

Inertia of the pulley is $0.5 MR^2 = I_p$

The total inertia of the system is $mR^2 + I_p$

11) A uniform solid cylinder of mass 10 kg can rotate about a frictionless axle through its center O, as shown in the cross-sectional view in the figure. A rope wrapped around the outer radius $R_1 = 1.0$ m exerts a force of magnitude $F_1 = 5.0$ N to the right. A second rope wrapped around another section of radius $R_2 = 0.50$ m exerts a force of magnitude $F_2 = 6.0$ N downward. What is the angular acceleration of the cylinder?





1- If the pulley has no mass.

Note: Mg is not equal to the tension
Because down (Mg) > up (tension)

Say $g = 10$.

We can have one single system

2 blocks $\rightarrow$

$$350 - 250 = (35 + 25) \times a$$

If we consider just one block $\rightarrow$

$$350 - T = 35a \quad \text{or}$$

$$T - 250 = 25a$$

$T_1 = T_2 = T$ because the pulley has no mass

2- If the pulley has mass and inertia I_p

$a = R(\alpha)$. $I_p = 0.5 MR^2$ usually

A) If we consider one single system:

(easier). Outside forces = 350N and 250 N

$$350R - 250R = (I_p + 35R^2 + 25R^2) \alpha$$

B) To find T_1

$$350 - T_1 = 35 \times a$$

With $a = R \alpha$

C) to find T_2

$$T_2 - 250 = 25a$$

If the pulley has mass $\rightarrow$

T_1 is not equal to T_2

Because T_1 fight against the inertia
Of the pulley.

Example: $M = 5\text{kg}$

$$I_p = 0.5 M R^2$$

Find a and α

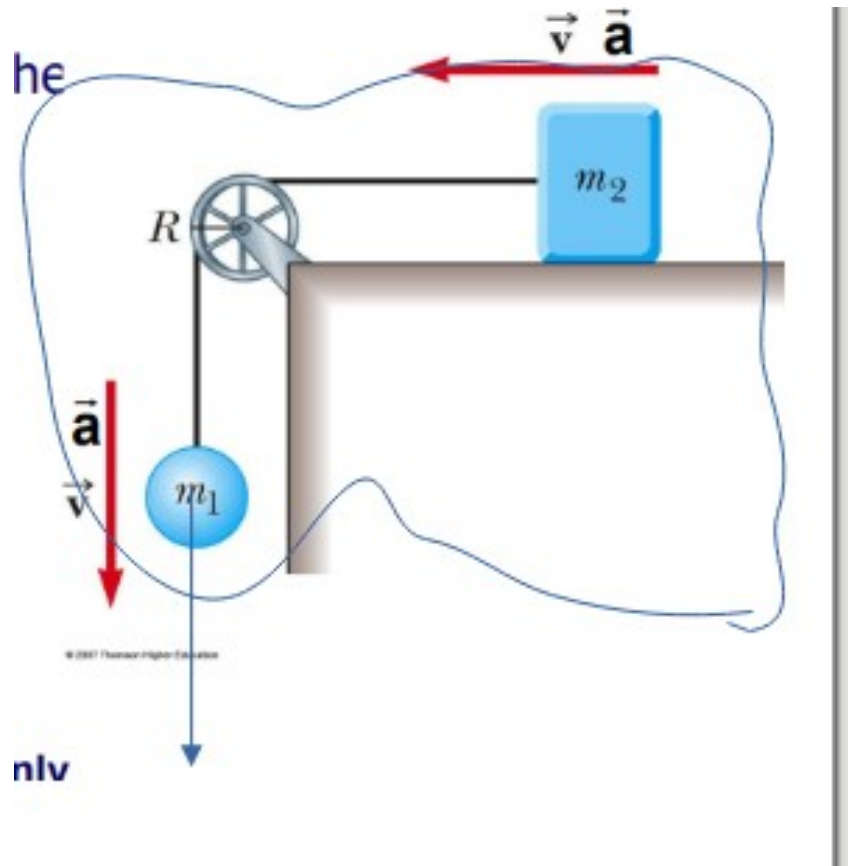
A wheel that is free to rotate has a radius of 15 centimeters. It has a cord wrapped around its circumference, and a 1.2-kilogram mass is attached to the lower end of the cord. If the mass is released, it falls a distance of 2.5 meters in 3.5 seconds. What is the moment of inertia of the wheel?

Use kinematics to find the acceleration a ($h = 0.5 a t^2$) Then $\alpha = a/R$
 $mgR = I \alpha$

A disk-shaped 18-kilogram flywheel has a radius of 20 centimeters and is spinning at 150 radians per second. A brake applied to its rim brings it to rest in 12 seconds. How much braking force was used?

Use kinematics to find the acceleration α ($\alpha = (w_f - w_i) / \text{time}$)
Then force $\times R = I \alpha$ with $I = MR^2$

Use the same principle



Consider the system inside the blue ribbon. The only outside force is m_1g
 The weight of m_2 is balanced by the normal. Does not contribute to motion.
 Imagine the masses are close to pulley. $I_{total} = I_p + m_1R^2 + m_2R^2$
 (Torque due to m_1g) = (total inertia) x (angular acceleration). EASY

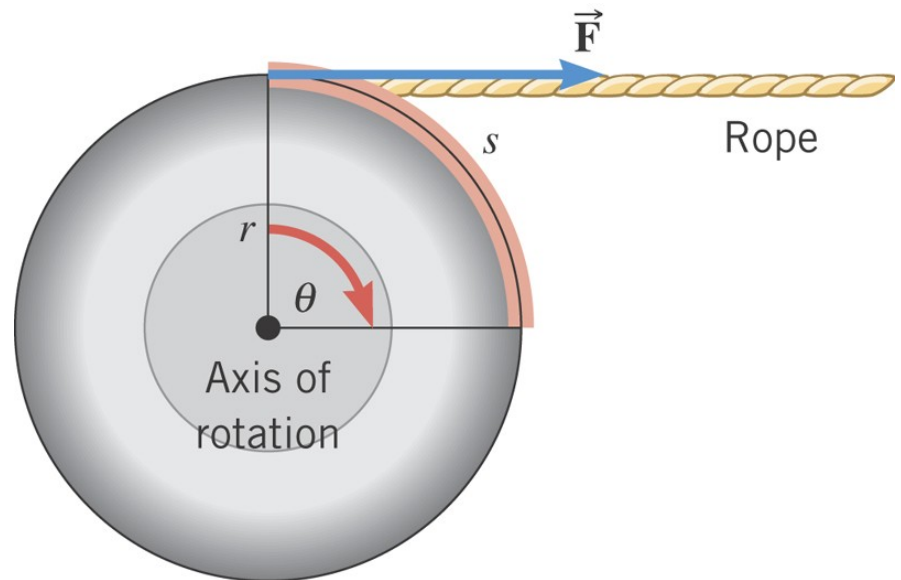
Rotational Work and Energy

$$s = r\theta$$

$$W = Fs = Fr\theta$$

$$\tau = Fr$$

$$W = \tau\theta$$



9.5 Rotational Work and Energy

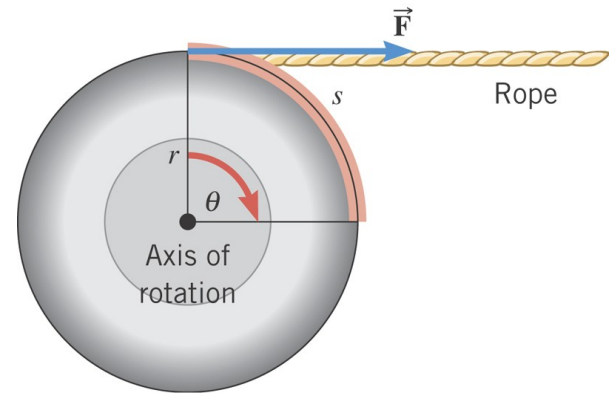
DEFINITION OF ROTATIONAL WORK

The rotational work done by a constant torque in turning an object through an angle is

$$W_R = \tau \theta$$

Requirement: The angle must be expressed in radians.

SI Unit of Rotational Work: joule (J)

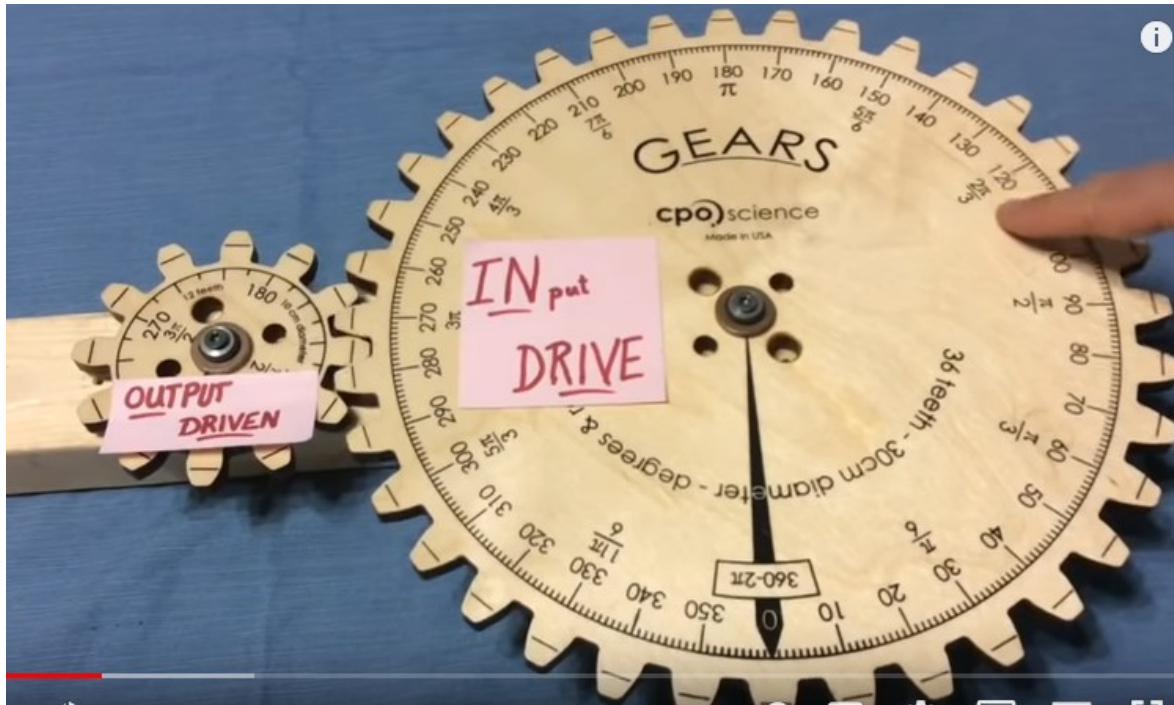


POWER for rotational motion = Work/time = torque x (angular speed)
Like Power = force x (linear speed)

We can explain gears with conservation of energy

Work in = work out.

small angle x big torque = large angle x small torque



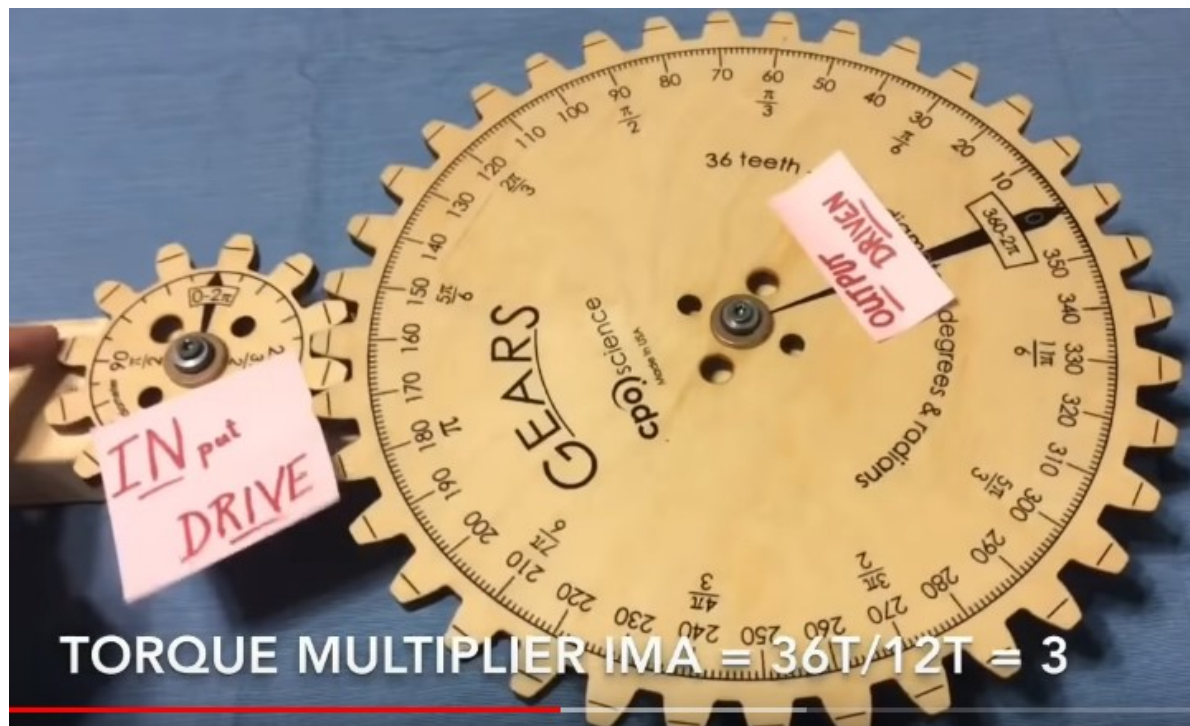
This is a speed Multiplier.

The MA is smaller than 1

$\text{Torque}_1 \times \text{angle}_1 =$
 $\text{Torque}_2 \times \text{angle}_2$

$\text{torque}_2 / \text{torque}_1 =$
 $\text{angle}_2 / \text{angle}_1 =$
 $\# \text{ of teeth}_2 / \# \text{ teeth}_1$

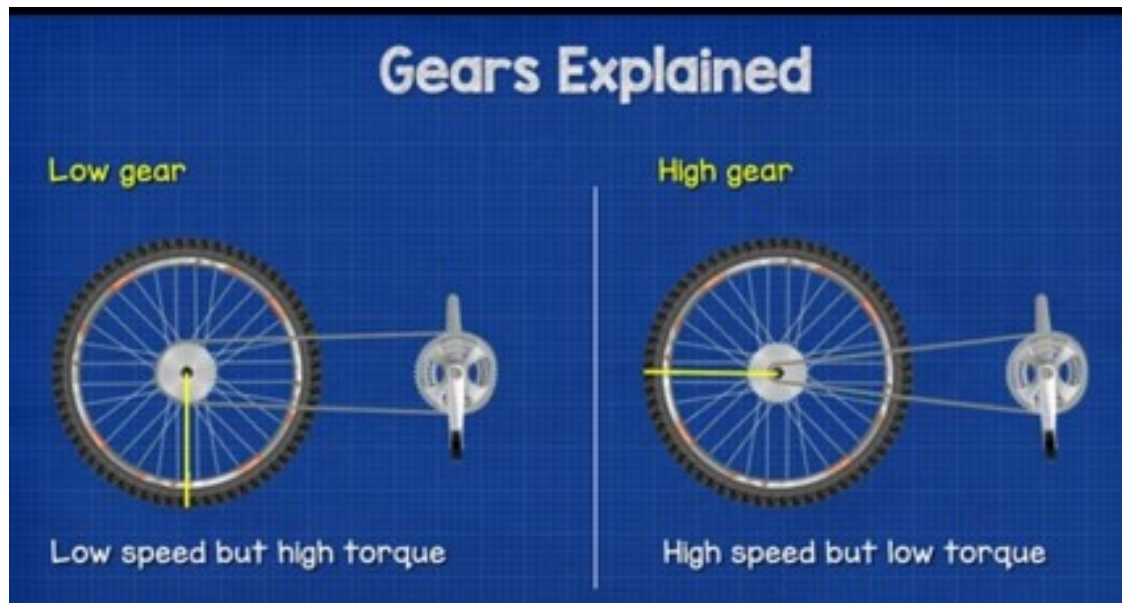
<https://www.youtube.com/watch?v=uz436lxb1-I>



This is a torque multiplier
= bike uphill

The MA is larger than 1

$F_1 \times R_1 \times$

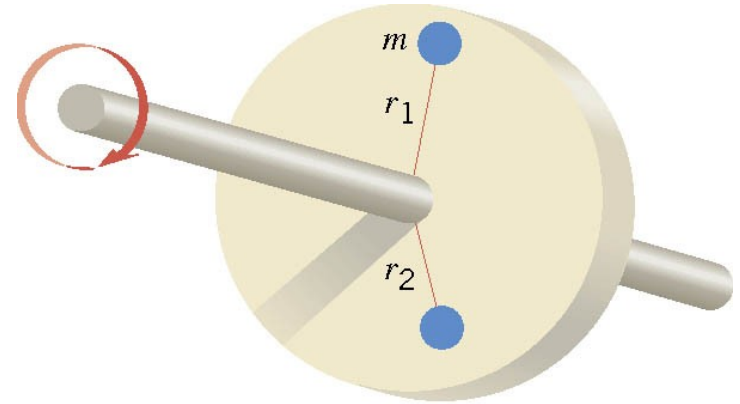


9.5 Rotational Work and Energy

$$KE = \frac{1}{2} m v_T^2 = \frac{1}{2} m r^2 \omega^2$$



$$v_T = r\omega$$



$$KE = \sum \left(\frac{1}{2} m r^2 \omega^2 \right) = \frac{1}{2} \left(\sum m r^2 \right) \omega^2 = \frac{1}{2} I \omega^2$$

9.5 Rotational Work and Energy

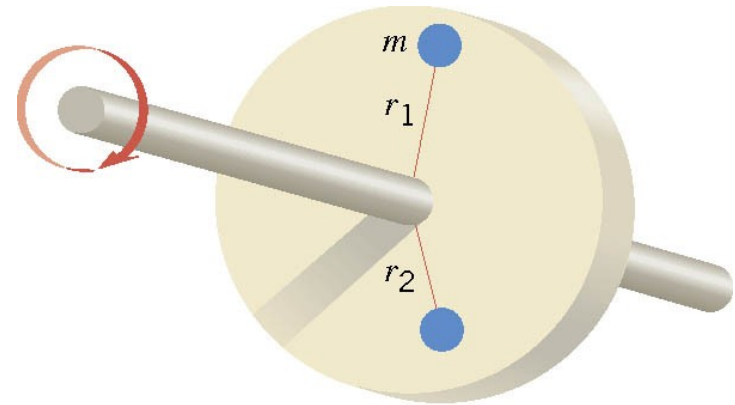
DEFINITION OF ROTATIONAL KINETIC ENERGY

The rotational kinetic energy of a rigid rotating object is

$$KE_R = \frac{1}{2} I \omega^2$$

Requirement: The angular speed must be expressed in rad/s.

SI Unit of Rotational Kinetic Energy: joule (J)



10) In a laboratory experiment, a student brings up the rotational speed of a flywheel to 30.0 rpm. She then allows it to slow down on its own, and counts 240 revolutions before the apparatus uniformly comes to a stop. The moment of inertia of the flywheel is $0.0850 \text{ kg} \cdot \text{m}^2$. What is the magnitude of the retarding torque on the flywheel?

- ✔ A) $0.000278 \text{ N} \cdot \text{m}$
- B) $0.0000136 \text{ N} \cdot \text{m}$
- C) $0.159 \text{ N} \cdot \text{m}$
- D) $0.0787 \text{ N} \cdot \text{m}$
- E) $0.0425 \text{ N} \cdot \text{m}$

Work-KE theorem

Work done by friction = change in KE

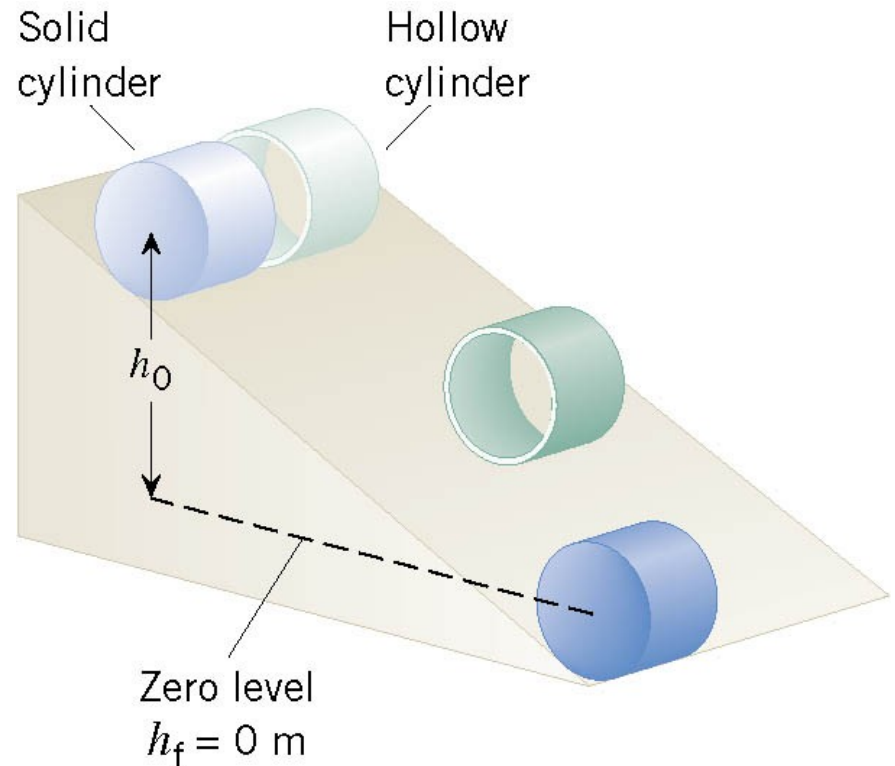
$$(\text{Torque by friction}) \times (\text{angular displacement}) = 0.5 I (\omega_f^2 - \omega_i^2)$$

$\omega_f = 0$ (it stops). Convert to radians

Example 13 Rolling Cylinders

A thin-walled hollow cylinder (mass = m_h , radius = r_h) and a solid cylinder (mass = m_s , radius = r_s) start from rest at the top of an incline.

Determine which cylinder has the greatest translational speed upon reaching the bottom.



9.5 Rotational Work and Energy

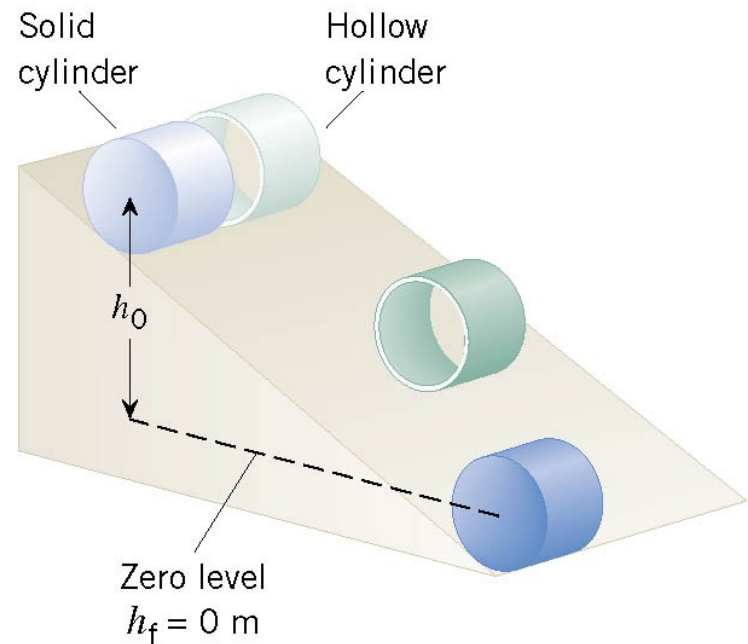
$$E = \frac{1}{2} m v^2 + \frac{1}{2} I \omega^2 + m g h$$

ENERGY CONSERVATION

$$\frac{1}{2} m v_f^2 + \frac{1}{2} I \omega_f^2 + m g h_f = \frac{1}{2} m v_i^2 + \frac{1}{2} I \omega_i^2 + m g h_i$$

$$\frac{1}{2} m v_f^2 + \frac{1}{2} I \omega_f^2 = m g h_i$$

$$\omega_f = v_f / r$$



9.5 Rotational Work and Energy

$$\frac{1}{2} m v_f^2 + \frac{1}{2} I v_f^2 / r^2 = m g h_i$$

$$v_f = \sqrt{\frac{2 m g h_o}{m + I / r^2}}$$

The cylinder with the smaller moment of inertia will have a greater final translational speed.

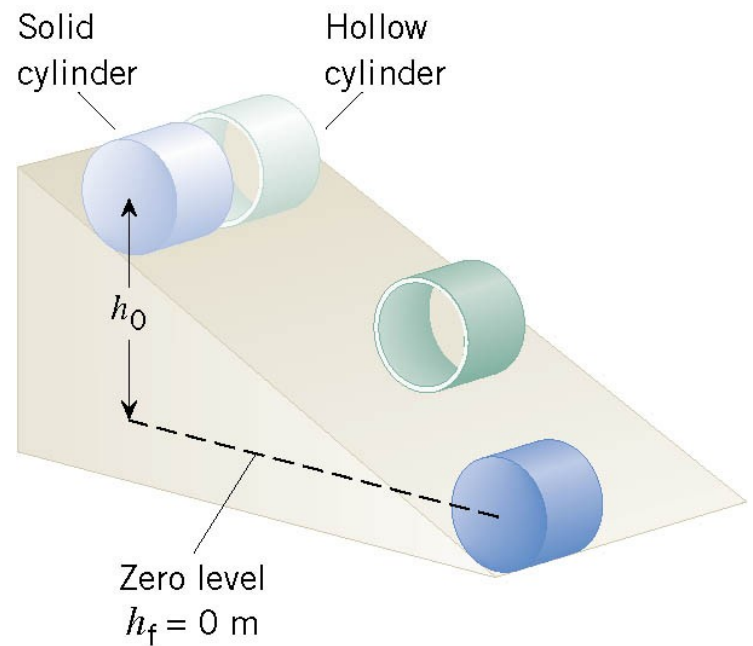
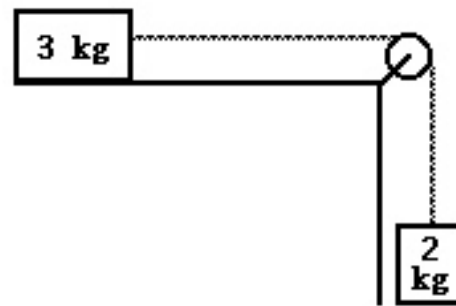


Figure 9.2



9) In Figure 9.2, two blocks, of masses 2 kg and 3 kg, are connected by a light string which passes over a pulley of moment of inertia $0.004 \text{ kg} \cdot \text{m}^2$ and radius 5 cms. The coefficient of friction for the table top is 0.30. The blocks are released from rest. Using energy methods, one can deduce that after the upper block has moved 0.6 m, its speed is: (see video)

9) _____

Potential energy of the 2kg – energy lost (friction) = total kinetic energy (with pulley)

$2 \times g \times 0.6 - \text{work friction} = \text{linear KE of 2kg} + \text{linear KE of 3kg} + \text{rotational KE pulley}$

Linear KE = $0.5 m V^2$ and rotation KE = $0.5 I \omega^2$ with $V = R \omega$

9.6 *Angular Momentum*

DEFINITION OF ANGULAR MOMENTUM

The angular momentum L of a body rotating about a fixed axis is the product of the body's moment of inertia and its angular velocity with respect to that axis:

$$L = I\omega$$

Requirement: The angular speed must be expressed in rad/s.

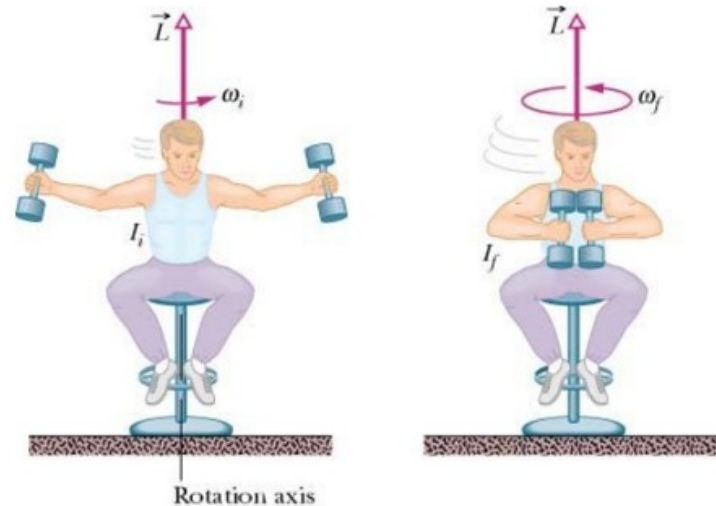
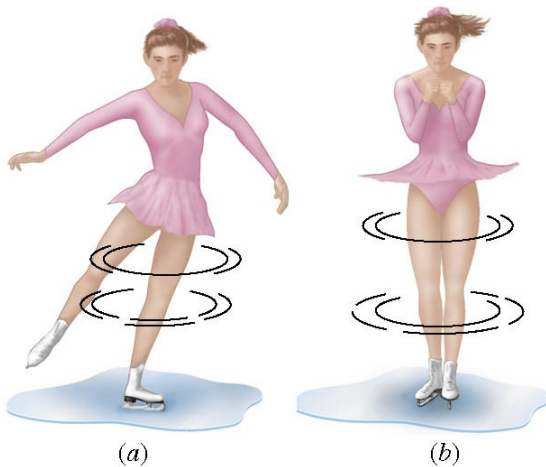
SI Unit of Angular Momentum: $\text{kg}\cdot\text{m}^2/\text{s}$

This is a special case of a symmetric object. The axis goes through the CM
The angular momentum is an intrinsic property of the system.
Otherwise, the angular momentum depends on a reference point/ (point of origin)

9.6 Angular Momentum

PRINCIPLE OF CONSERVATION OF ANGULAR MOMENTUM

The angular momentum of a system remains constant (is conserved) if the net external torque acting on the system is zero.

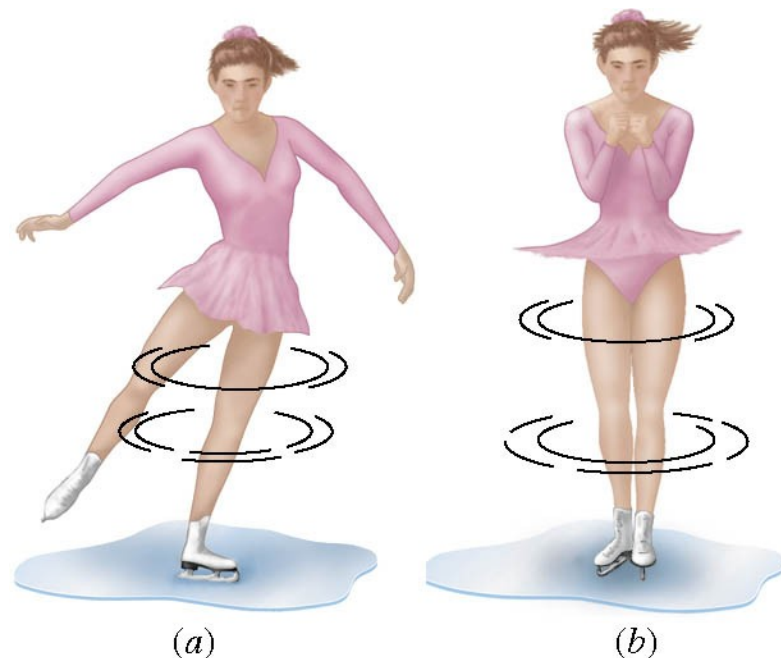


**Isolated
System**

Conceptual Example 14 A Spinning Skater

An ice skater is spinning with both arms and a leg outstretched. She pulls her arms and leg inward and her spinning motion changes dramatically.

Use the principle of conservation of angular momentum to explain how and why her spinning motion changes.

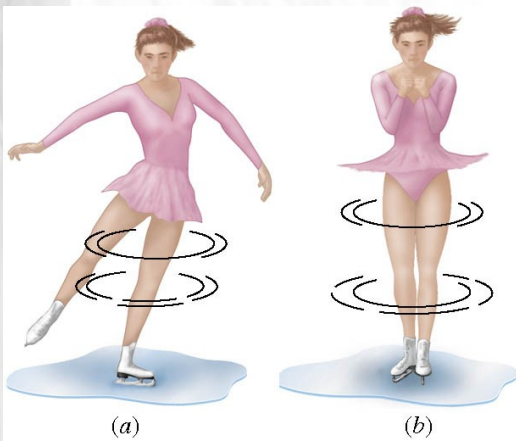


Video science theater
Neutron star

<https://www.youtube.com/watch?v=5CZBBbnVmUA> head spin

<http://www.youtube.com/watch?v=0k276y9kuQQ&feature=related>

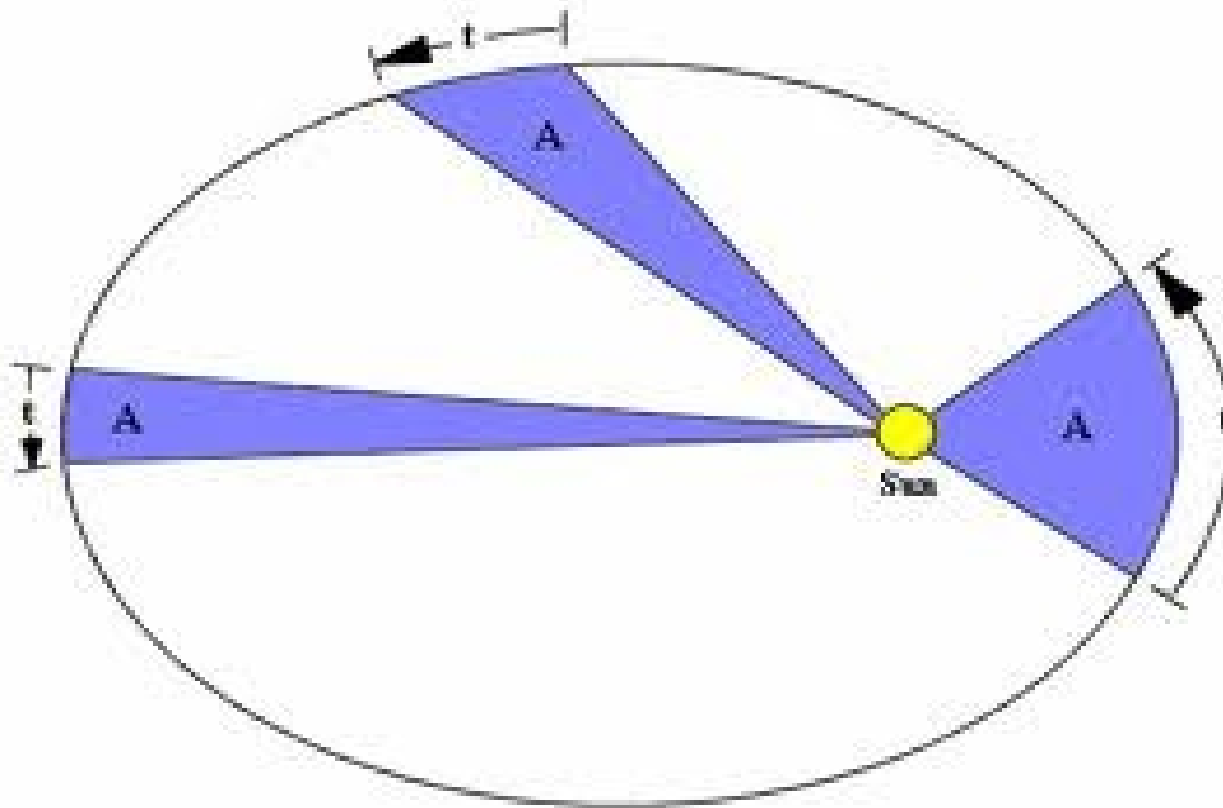
Simple demo



$$I_1 \omega_1 = I_2 \omega_2$$



Video astro

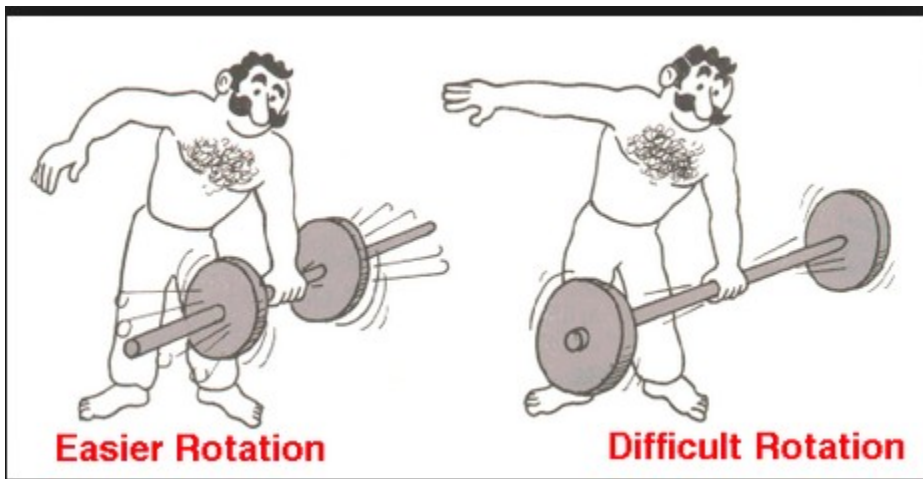


http://highered.mcgraw-hill.com/olcweb/cgi/pluginpop.cgi?it=swf::800::600::/sites/dl/free/0072482621/78778/Kepler_Nav.swf::Keplers%20Second%20Law%20Interactive

Kepler's second law: the planets sweep the same area during
The same amount of time.
It is a consequence of conservation of angular momentum.

CONSERVATION OF ANGULAR MOMENTUM

Angular momentum depends on the mass distribution about an axis and On the speed. If the distribution of mass is modified, the speed has to Change so the momentum stays the same.



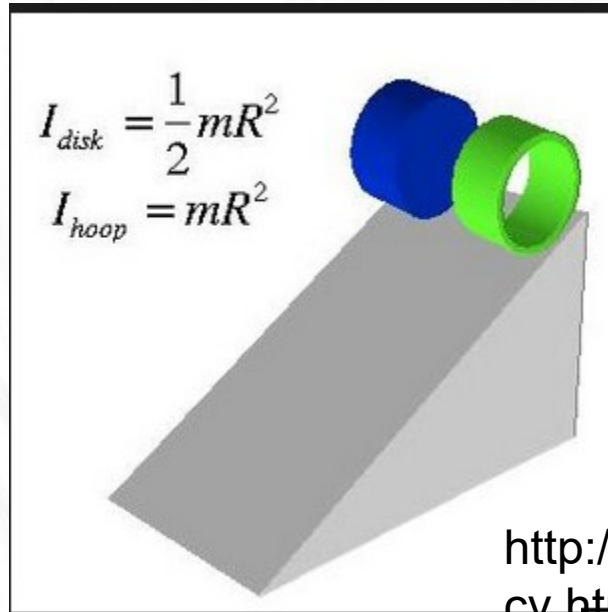
This happens during EQ or tsunami.

2010 a monster quake in Chile shortened the day by 1.26 millionths of A second so the Earth spins (slightly faster)

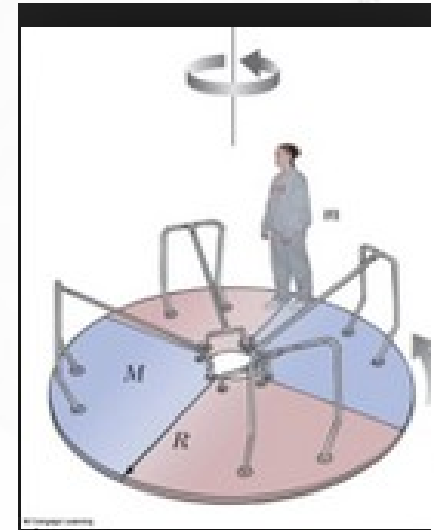


Town of Iloca 168 miles of Santiago. 15 story building

<https://www.youtube.com/watch?v=tcs93mPn91E>



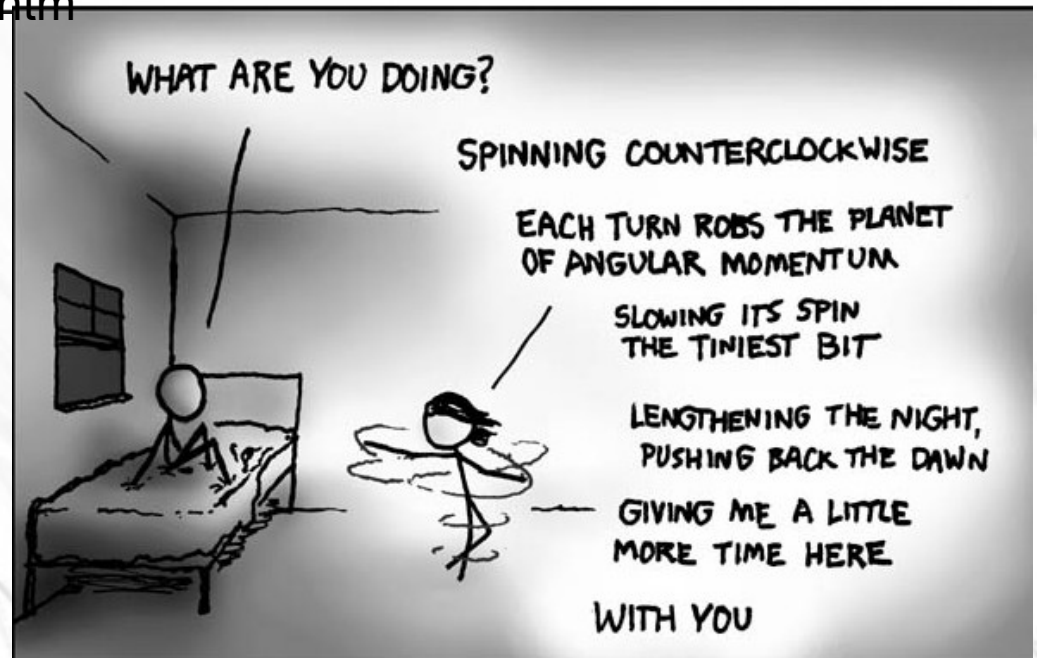
Which one wins ?

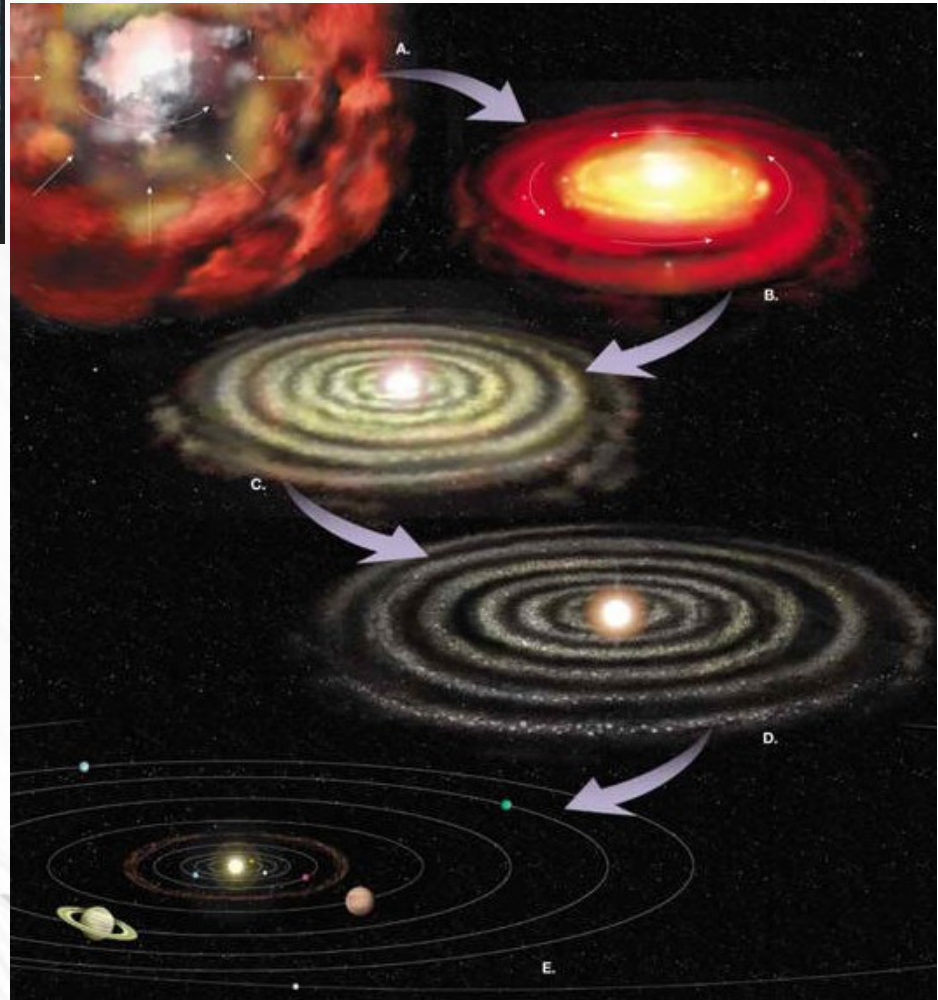


<http://www.personal.psu.edu/djs75/cv.htm>



© Veronique Lankar 2017

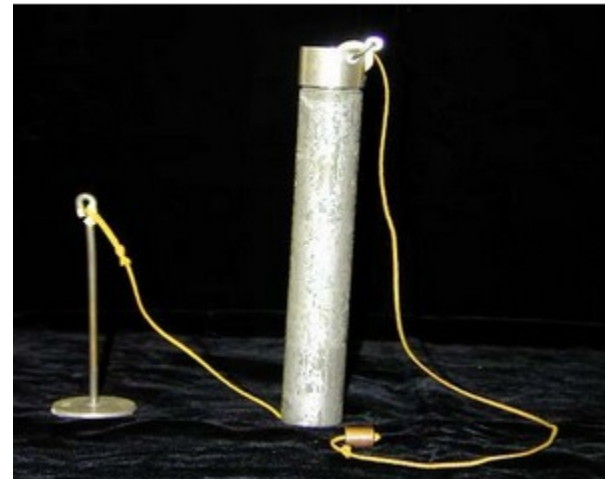
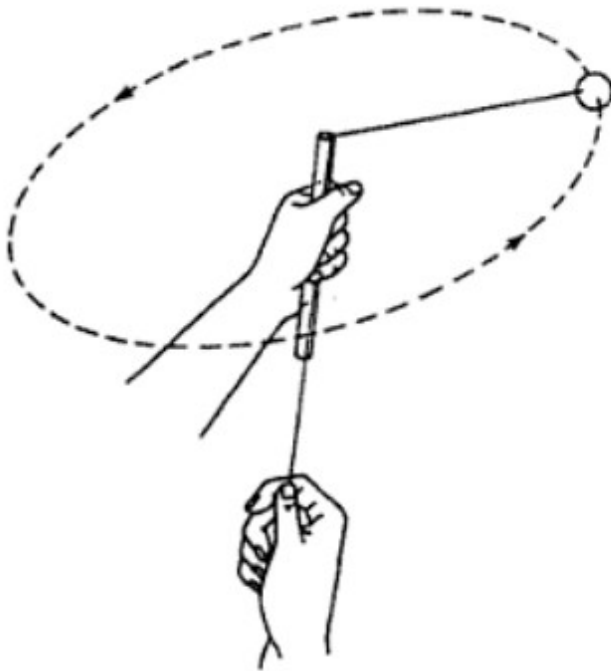




The rotating string in Figure 4-3 is 1.5 meters long, and the ball is going 22 meters per second. How fast will it be going if the string is pulled down until the radius of the circle is reduced to 80 centimeters?

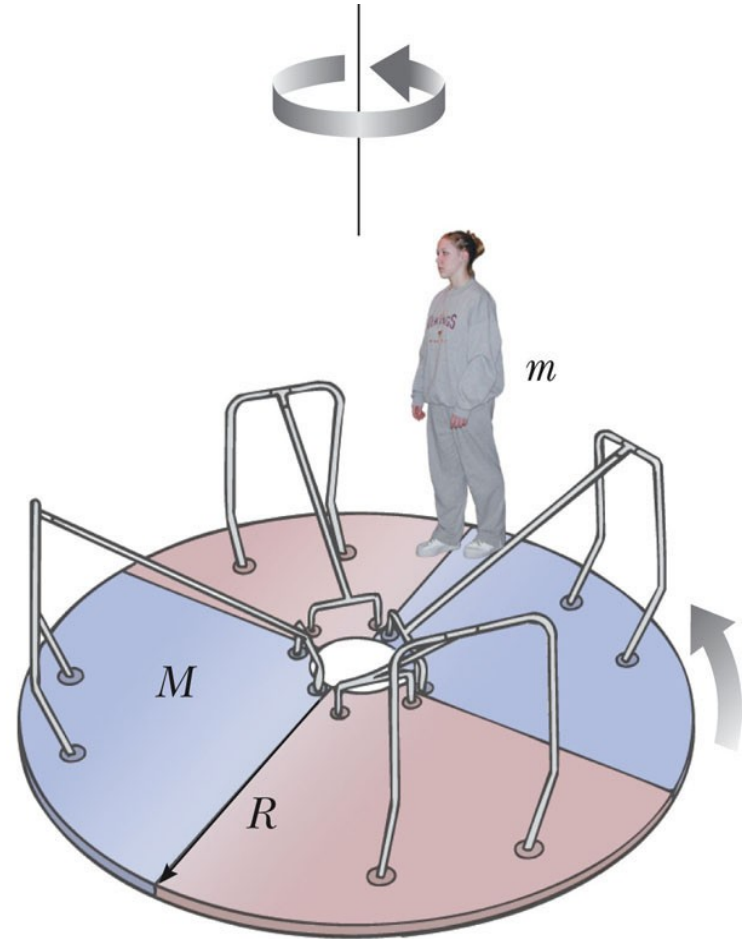
Solution:

$$\text{Use } R_1 V_1 = R_2 V_2$$



The Merry-Go-Round

- The moment of inertia of the **system** = the moment of inertia of the platform plus the moment of inertia of the person.
 - Assume the person can be treated as a particle
- As the person moves toward the center of the rotating platform the moment of inertia decreases.
- The angular speed must increase since the angular momentum is constant.



Example: A merry-go-round problem

A 40-kg child running at 4.0 m/s jumps tangentially onto a stationary circular merry-go-round platform whose radius is 2.0 m and whose moment of inertia is 20 kg-m². There is no friction.

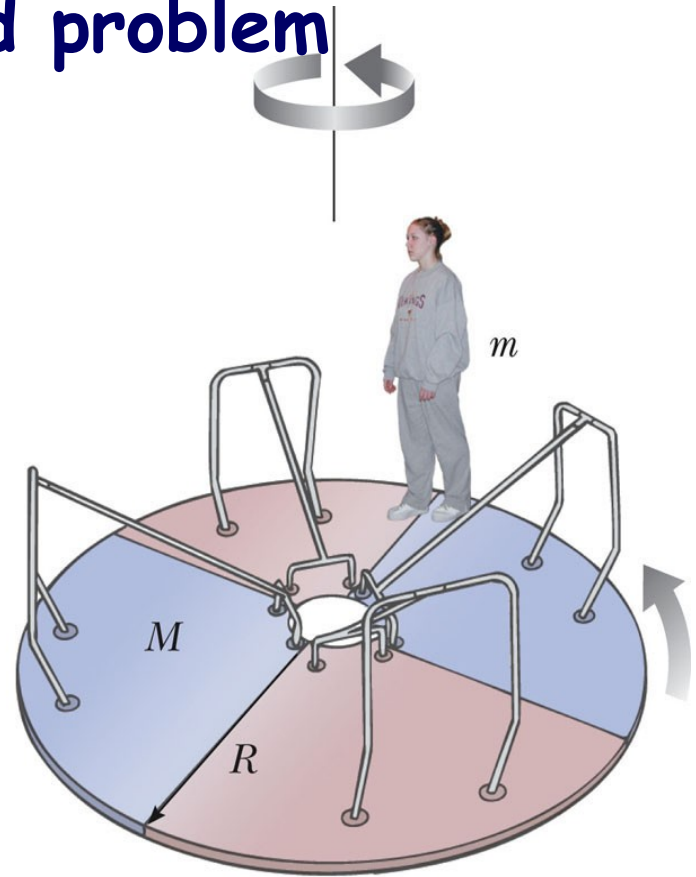
Find the angular velocity of the platform after the child has jumped on.

You can use
Momentum before = momentum after
It's a collision.

Before $L_{\text{child/axis}} = mV_1 R_1$ (or $mR_1^2 \omega_1$ with $V = R_1 \omega_1$)

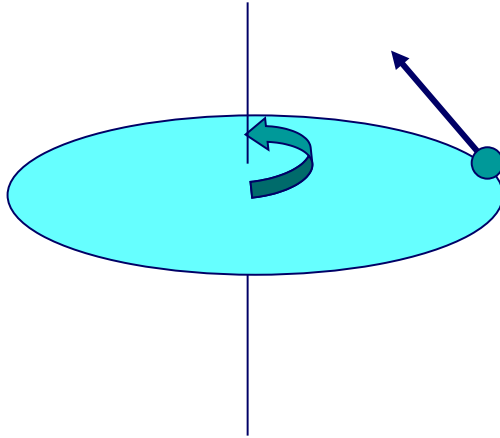
After $L_{\text{child}} + L_{\text{merry}} / \text{axis} = (mR_1^2 + I) \omega_2$

The system is isolated $\rightarrow$ momentum is conserved



© 2007 Thomson Higher Education

Solution: A merry-go-round problem



$$\mathbf{L}_{tot} = \sum I_i \boldsymbol{\omega}_i = \sum I_f \boldsymbol{\omega}_f$$

$$L_i = I_i \omega_i = I \omega_0 + m_c v_T r = m_c v_T r$$

$$L_f = I_f \omega_f = (I + m_c r^2) \omega_f$$

$$(I + m_c r^2) \omega_f = m_c v_T r$$

$$I = 20 \text{ kg.m}^2$$

$$V_T = 4.0 \text{ m/s}$$

$$m_c = 40 \text{ kg}$$

$$r = 2.0 \text{ m}$$

$$\omega_0 = 0$$

$$\omega_f = \frac{m_c v_T r}{I + m_c r^2} = \frac{40 \times 4 \times 2}{10 + 40 \times 2^2} = 1.78 \text{ rad/s}$$

First Example

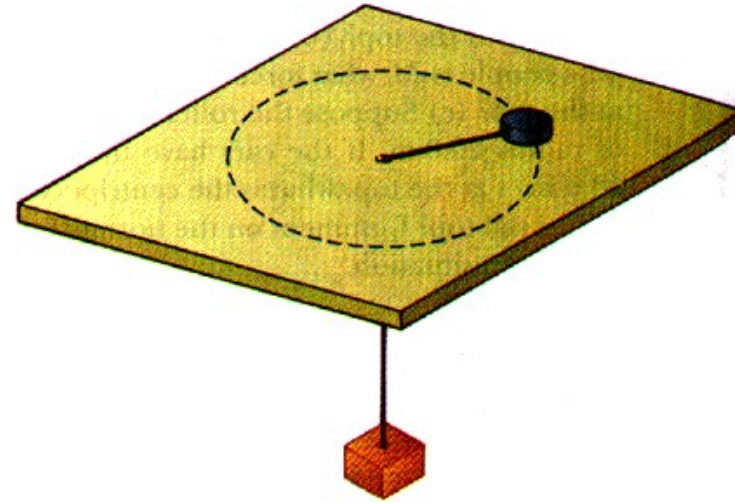
A puck of mass $m = 0.5 \text{ kg}$ is attached to a taut cord passing through a small hole in a frictionless, horizontal surface. The puck is initially orbiting with speed $v_i = 2 \text{ m/s}$ in a circle of radius

- ☐ $r_i = 0.2 \text{ m}$. The cord is then slowly pulled from below, decreasing the radius of the circle to $r = 0.1 \text{ m}$.
- ☐ What is the puck's speed at the smaller radius?
- ☐ Find the tension in the cord at the smaller radius.

The angular momentum is conserved because there is
No external torque relative to center.

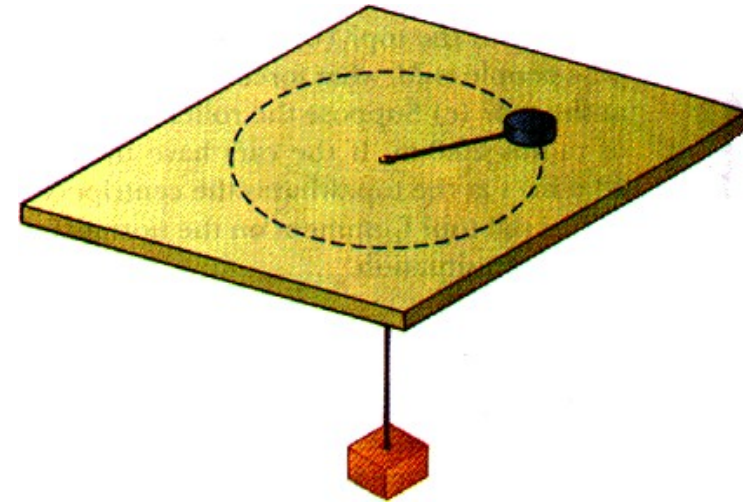
The torque from pull=0. $\rightarrow R_1 V_1 = R_2 V_2$

the centripetal force the tension = mV^2/R (slow means $T = F_c$)



Angular Momentum Conservation

- ❑ $m = 0.5 \text{ kg}$, $v_i = 2 \text{ m/s}$, $r_i = 0.2 \text{ m}$, $r_f = 0.1 \text{ m}$, $v_f = ?$
- ❑ Isolated system?
- ❑ Tension force on m exert zero torque about hole, why?



$$\mathbf{L}_i = \mathbf{L}_f \quad \mathbf{L} = \mathbf{r} \times \mathbf{p} = \mathbf{r} \times (m\mathbf{v})$$

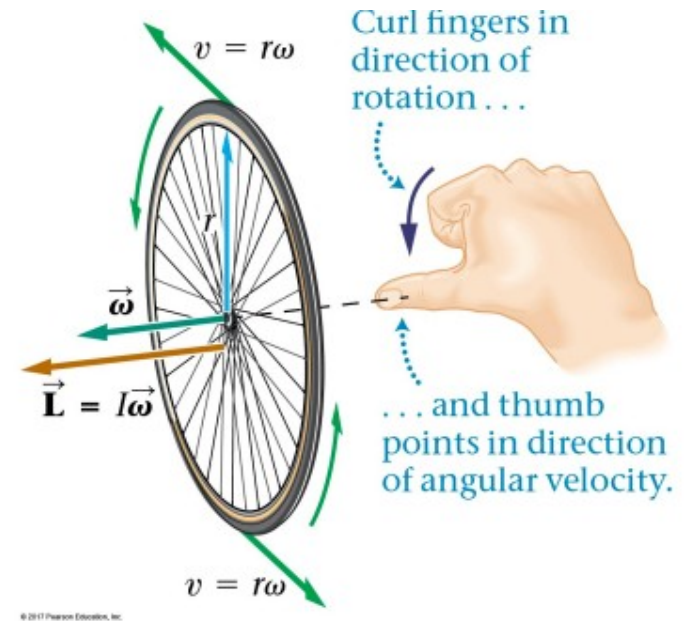
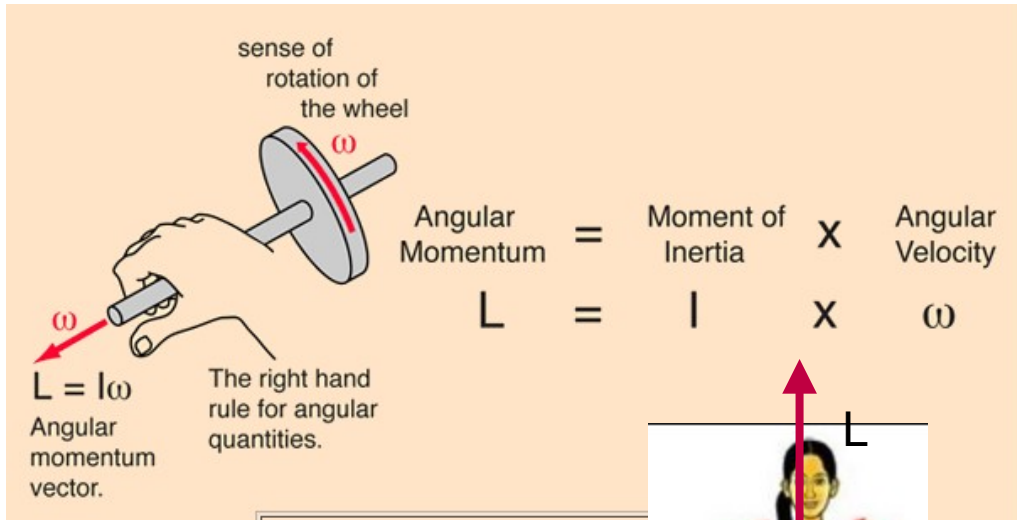
$$L_i = m r_i v_i \sin 90^\circ = m r_i v_i \quad L_f = m r_f v_f \sin 90^\circ = m r_f v_f$$

$$v_f = \frac{r_i}{r_f} v_i = \frac{0.2}{0.1} 2 = 4 \text{ m/s}$$

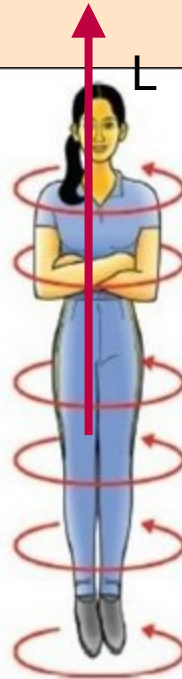
$$T = m a_c = m \frac{v_f^2}{r_f} = 0.5 \frac{4^2}{0.1} = 80 \text{ N}$$

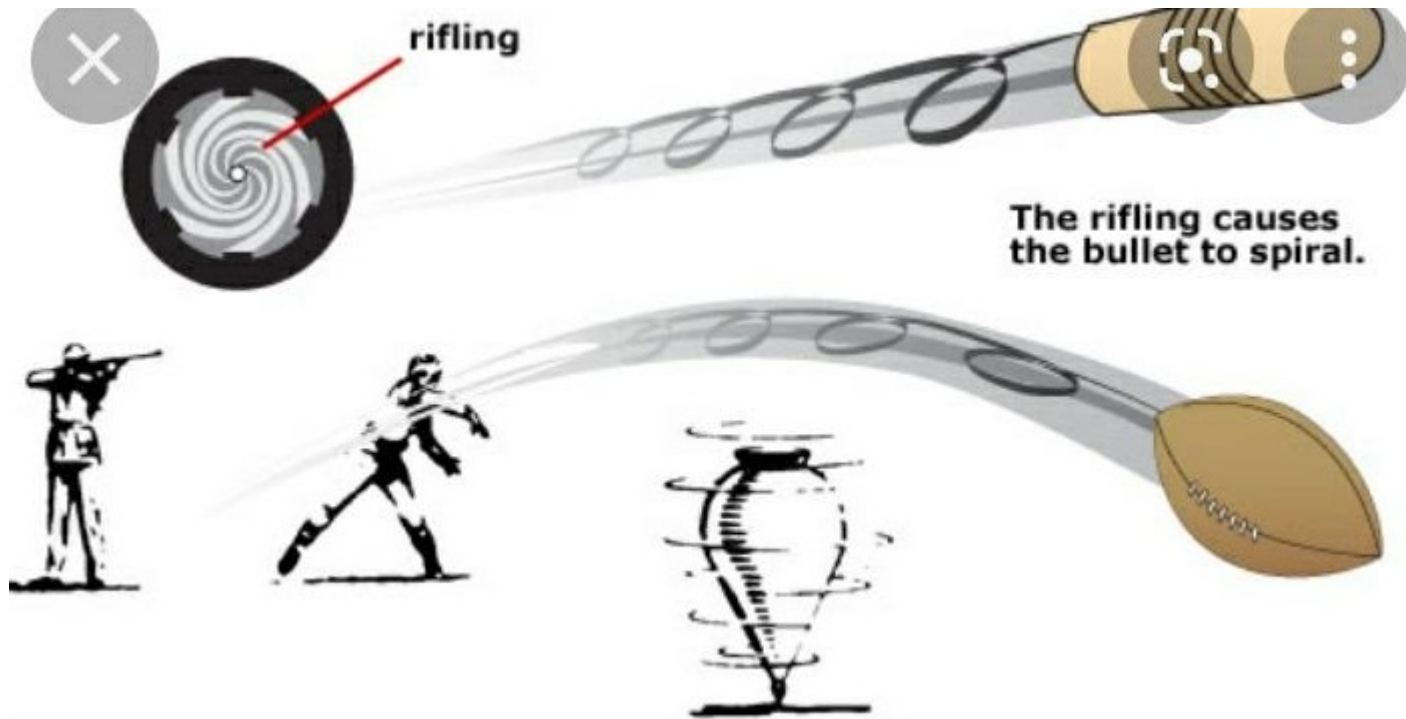
Angular momentum is a vector

If there is no external torque, the magnitude and direction stays the same. Direction is given by the right hand rule.



Me spinning
(and younger,,)





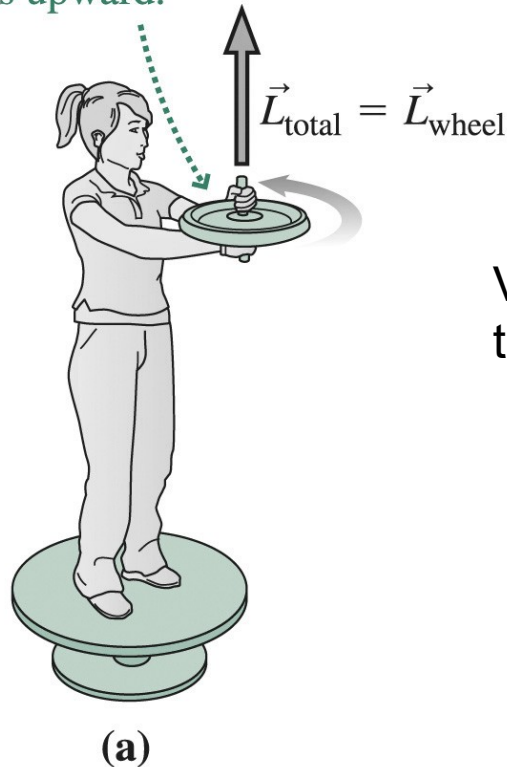
The vector angular momentum “wants” to point to the same direction
Unless there is an external force. If inertia is large (large mass, large I)
Or if the speed is large, a large torque is needed to destabilize the
Spinning objects. See the [video](#) gyroscope in space. →

A gyroscope is Used for orientation

Without external torque, momentum stays constant. Magnitude AND DIRECTION.

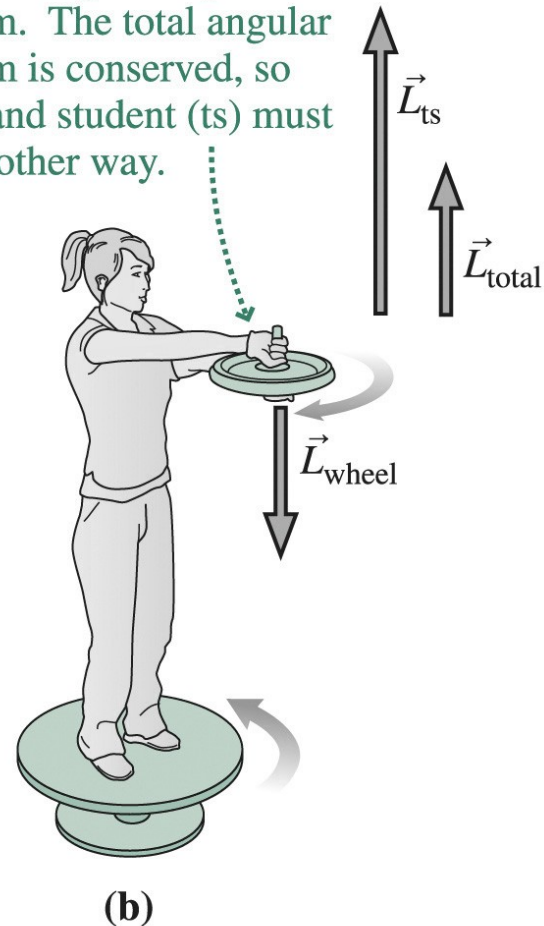
Here before momentum of the system (stool, girl, wheel) is up so without Friction it want to stay up. It the idea of reaction wheels (space, robotics) → managing momentum.

The student stands on a stationary turntable holding a wheel that spins counterclockwise; the wheel's angular momentum points upward.



Video science theater

She flips the spinning wheel, reversing its angular momentum. The total angular momentum is conserved, so turntable and student (ts) must rotate the other way.



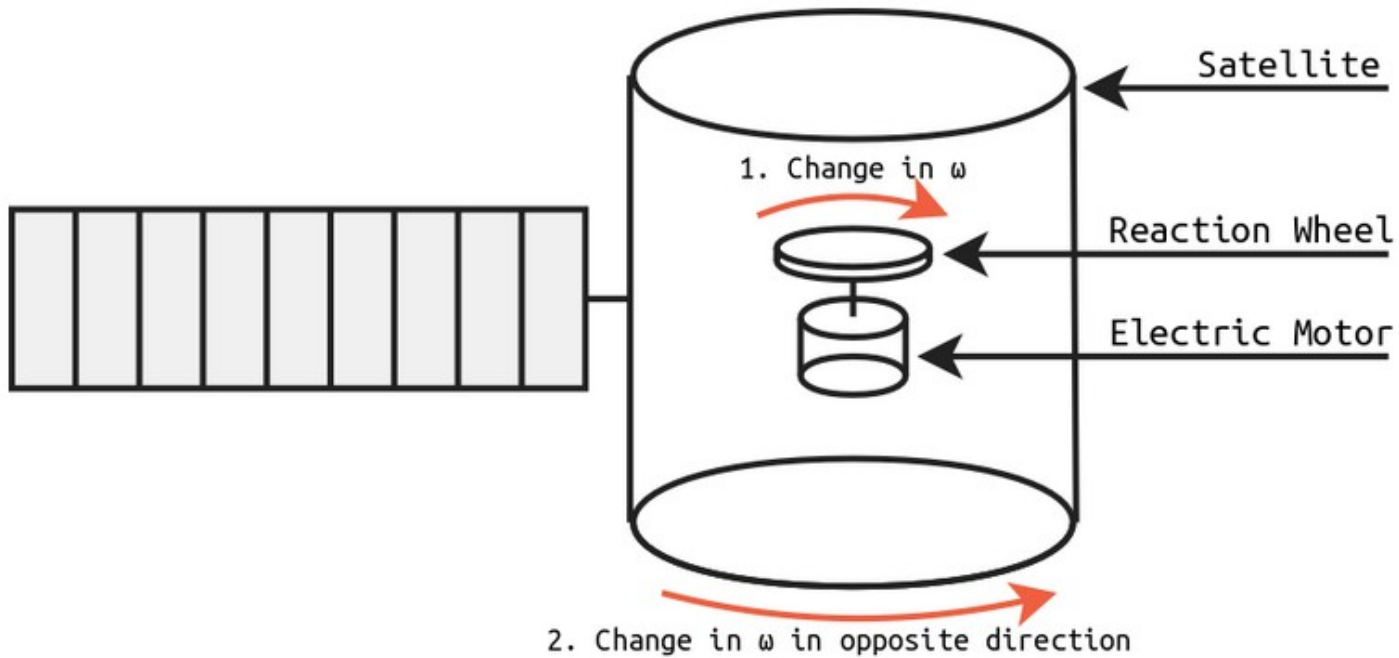


Diagram of a satellite with reaction wheels

<https://charleslabs.fr/en/project-Reaction+Wheel+Attitude+Control>

<https://learn.parallax.com/tutorials/robot/elev-8/understanding-physics-multirotor-flight/rotation-torque-and-angular-momentum>



The cat spins the tail in one direction
And the body spins the other way.

This is the conservation of momentum.

Same idea with astronauts.

<https://www.youtube.com/watch?v=RtWbpyjJqrU>

<http://www.physlink.com/education/askexperts/ae411.cfm>

Recall: Linear Momentum and Force

- Linear motion: apply force to a mass
- The force causes the linear momentum to change
- The net force acting on a body is the time rate of change of its linear momentum

$$\mathbf{F}_{\text{net}} = \Sigma \mathbf{F} = m\mathbf{a} = m \frac{d\mathbf{v}}{dt} = \frac{d\mathbf{p}}{dt}$$

$$\mathbf{p} = m\mathbf{v}$$

$$\mathbf{I} = \mathbf{F}_{\text{net}} \Delta t = \Delta \mathbf{p}$$

A force applied during a time $t \rightarrow$ changes the linear momentum

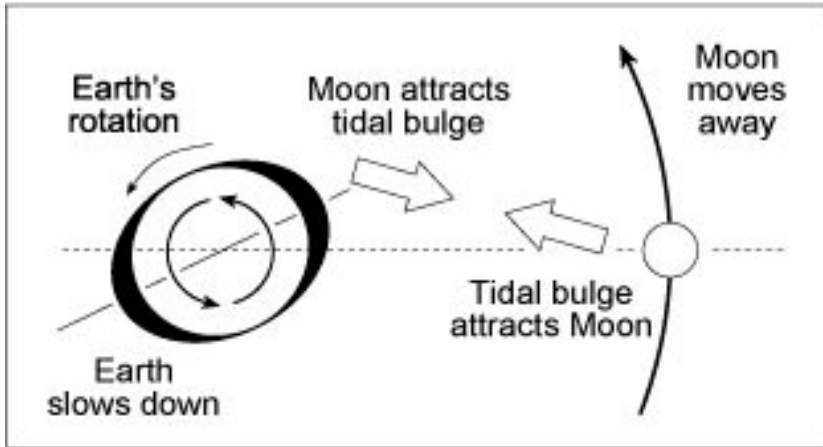
A torque applied during a time $t \rightarrow$ changes the angular momentum,

Angular Momentum and Torque

- Rotational motion: apply torque to a rigid body
- The torque causes the angular momentum to change
- The net torque acting on a body is the time rate of change of its angular momentum

$$\mathbf{F}_{\text{net}} = \Sigma \mathbf{F} = \frac{d\mathbf{p}}{dt} \quad \longrightarrow \quad \boldsymbol{\tau}_{\text{net}} = \Sigma \boldsymbol{\tau} = \frac{d\mathbf{L}}{dt}$$

- $\Sigma \boldsymbol{\tau}$ and $\mathbf{L}$ are to be measured about the same origin
- The origin must not be accelerating (must be an inertial frame)



Torque changes the angular momentum

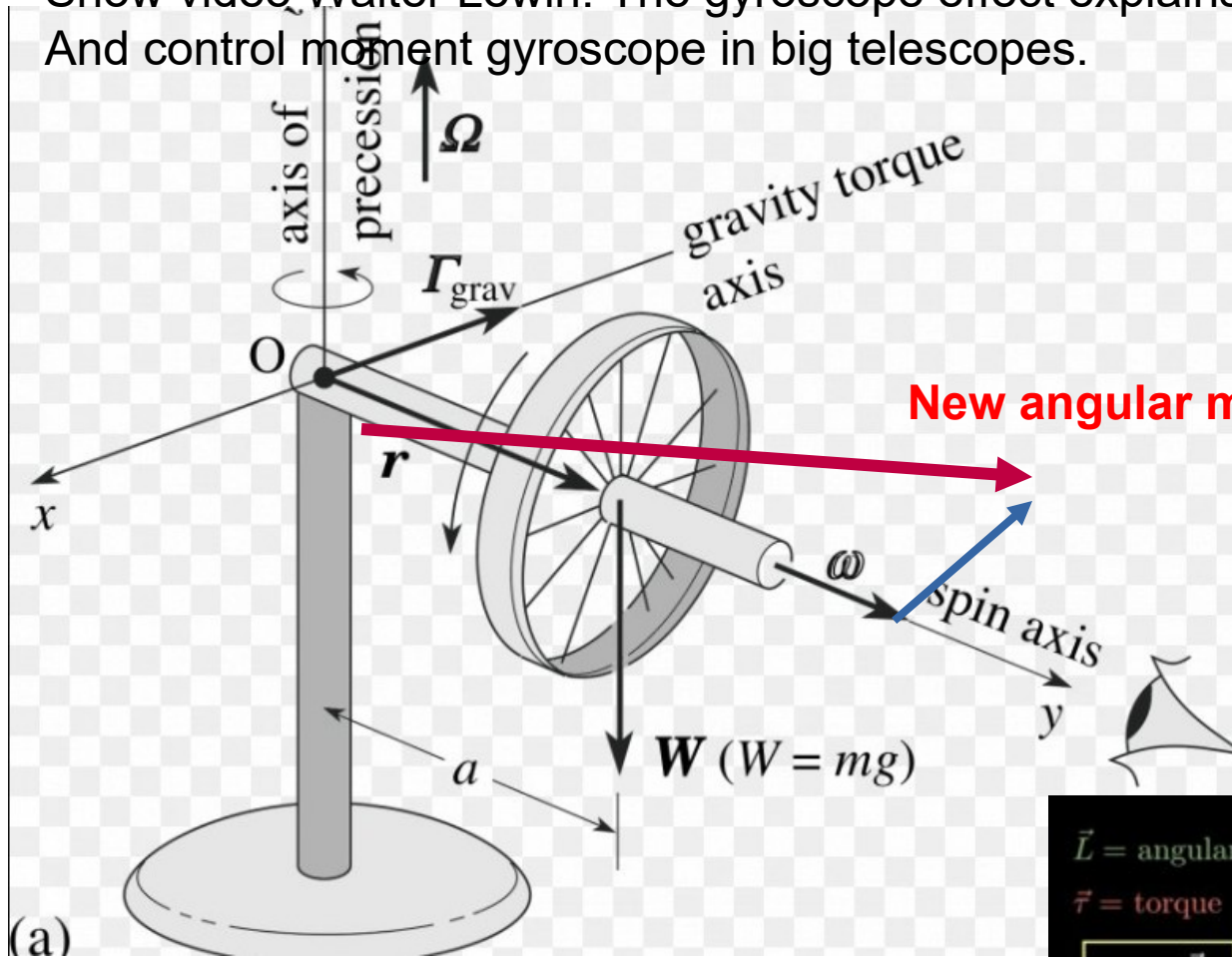
The torque that the Earth applies on the Moon changes the angular momentum of the Moon. The Moon gains momentum And the Earth loses momentum. The Moon moves away from us/

The Moon pulls on the bulges of the Earth (as a reaction to the pull from the Earth)
 The Earth loses spin momentum and the Moon gain orbital momentum
 (or you can see the bulges are ahead of Earth. There is a torque applied
 To the Moon that increases its orbital momentum)

L_{orbital} is proportional to mass and R $L = m v R = m R \sqrt{GM/R} = m \sqrt{GMR}$
 So if L increases (Moon) then R increases $\rightarrow$ Moon moves away from us

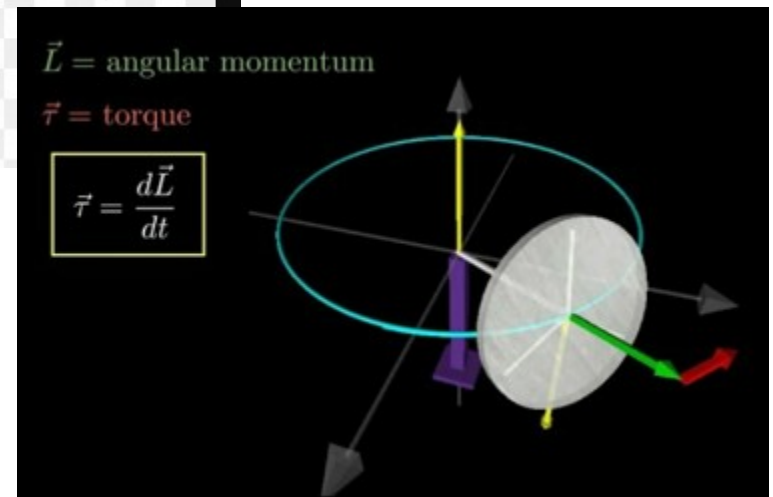
And larger $R \rightarrow$ slower orbital speed. So Farther and slower.
 The spin rotation of the Moon decreases too. As Moon and Earth are locked together.

Show video Walter Lewin. The gyroscope effect explains counter steering!
And control moment gyroscope in big telescopes.



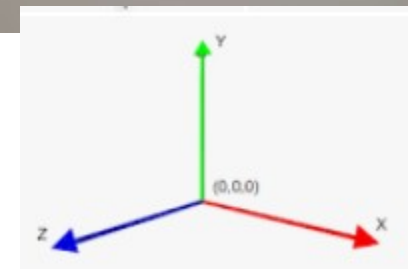
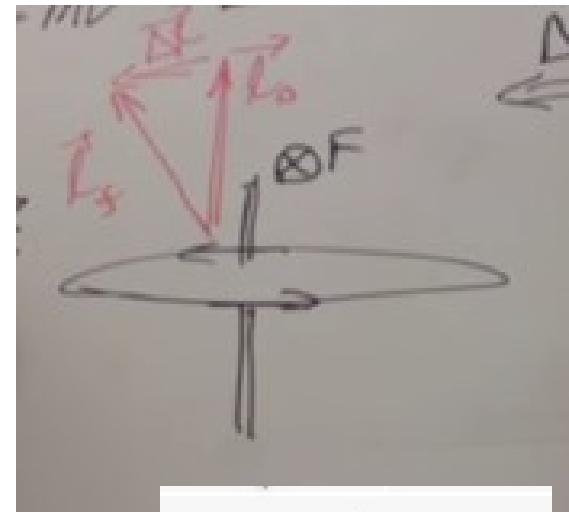
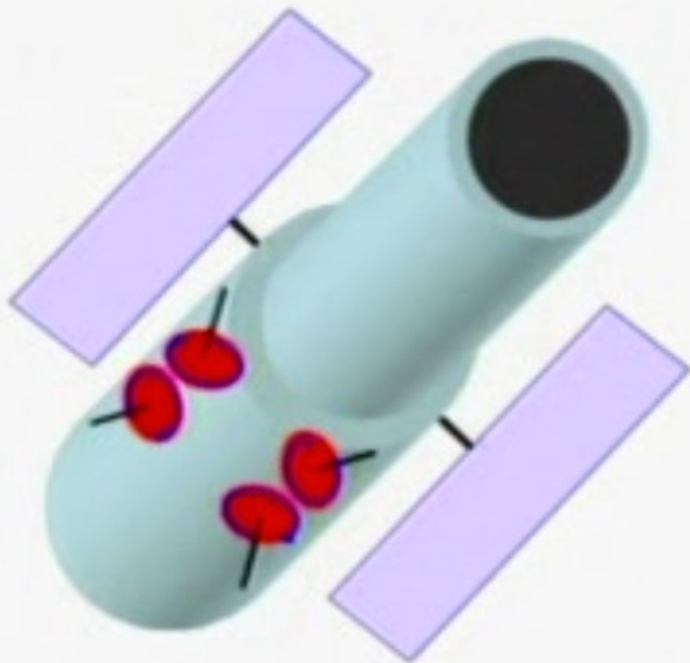
New angular momentum

Torque produced by
Weight $\rightarrow \Delta \mathbf{L}$



The Hubble Space Telescope is equipped with **reaction wheels** to change its overall orientation, rotating the entire telescope about the speed of a minute hand on a clock – 90 degrees in 15 minutes. But to stay pointed at a single target, it uses another technology: gyroscopes. Aug 13, 2019

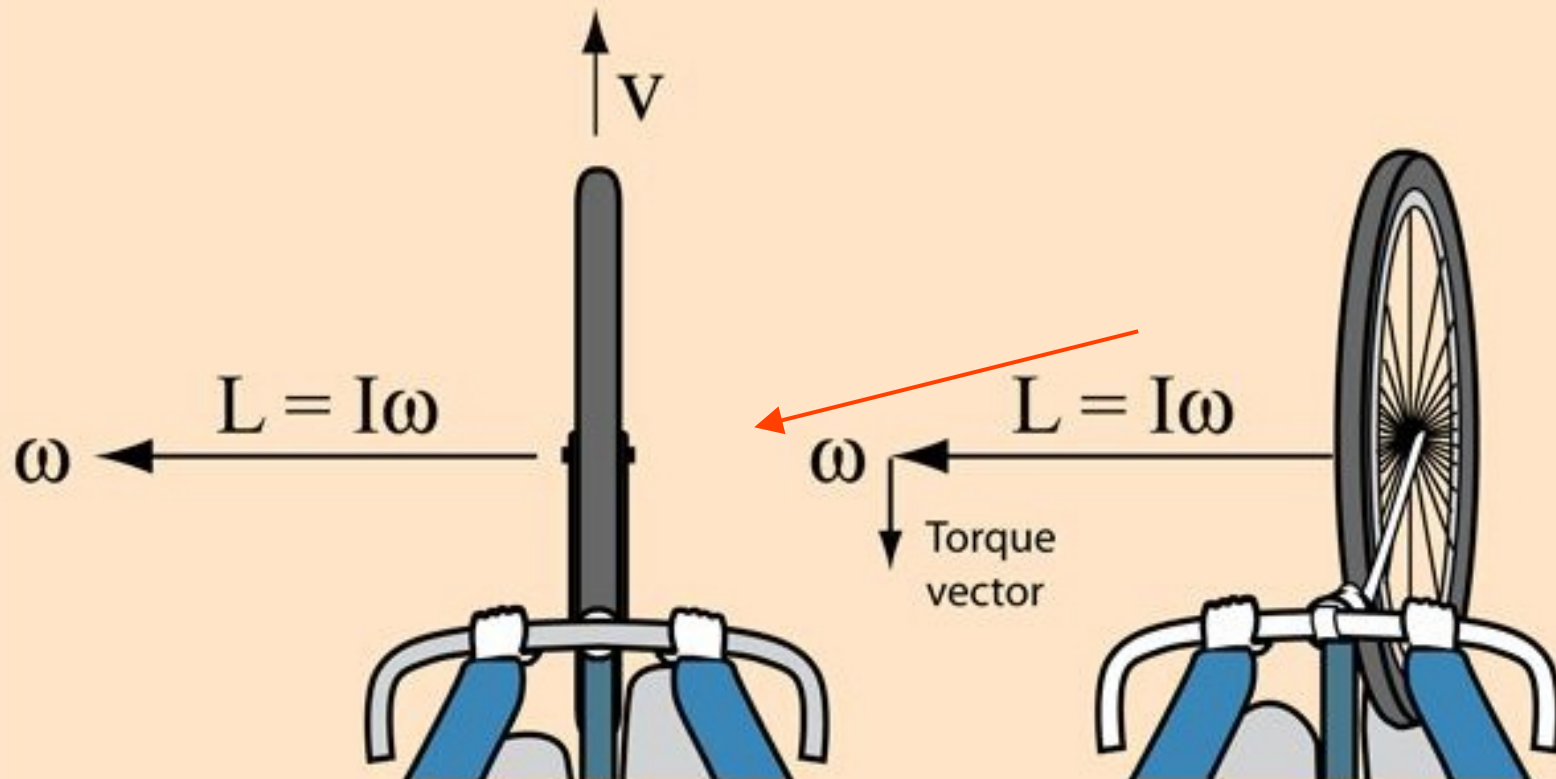
Hubble's Angular Momentum

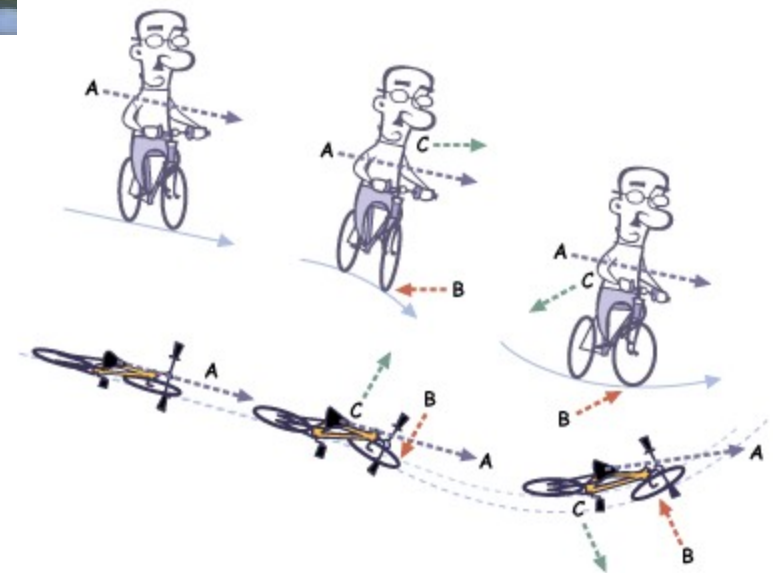
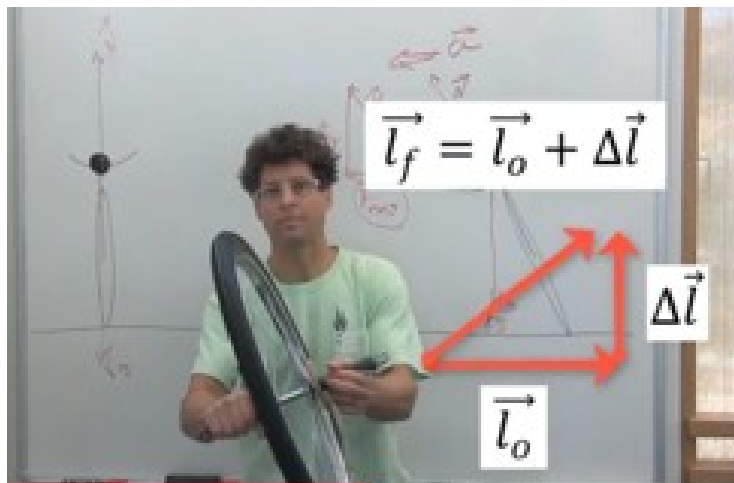


Rotation about the y-axis by force →
The wheel rotates about the zaxis

A bicycle held straight up will tend to go straight. It is tempting to say that it is stabilized by the gyroscopic action of the bicycle wheels, but the gyroscopic action is quite small.

If the rider leans left, a torque will be produced which causes a counterclockwise precession of the bicycle wheel, tending to turn the bicycle to the left.





<http://www.terrycolon.com/1features/counter-steering-easy.html>